

99 WAYS TO SURVIVE A ZOMBIE
APOCALYPSE

99 Ways to Survive a Zombie Apocalypse

*Ninety-nine rules collected from the people who wrote
them down, what each one is worth, and where it fails.*

MURTAZA RAZA

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CONTENTS

How to Use This Book	1
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PART I · THE FIRST WEEK

1	The First Hour	7
2	What You Believe	17
3	Moving	27

PART II · THE BODY AND THE BUILDING

4	Doors, and Where You Sleep	39
5	Noise	49
6	Water, Food, and Teeth	59

PART III · EVERYONE ELSE

7	Other People	71
8	The Group	81
9	The Long Middle	91

END MATTER

A Note on the Sources	103
Index of Names	107

BEFORE YOU BEGIN

How to Use This Book

This is a manual of ninety-nine rules for staying alive, assembled from the papers of people who for the most part did not.

Each entry has four parts. The rule, in the words somebody wrote it down in. Who kept it, what they did before, and what day they were on when they wrote it. Then two paragraphs that are mine: why the rule works, which is the reasoning, and how it goes wrong, which is the body.

The last paragraph is the point of the book. Anybody can tell you to keep quiet and stay off the roads. You know that. What nobody tells you is the precise circumstance in which keeping quiet gets your whole group killed, and there is one, and it is in chapter five. A rule with no stated failure is not a rule. It is a slogan. Slogans are most of why the first winter went the way it did.

The rules disagree with each other, and I have left them disagreeing. Rule 70 says the door does not open, ever, and the man who wrote it has buried people who opened theirs. Rule 77 was written by a woman who has opened hers every time she has been asked, for four years, and who is alive, and whose settlement is the largest in this valley. Both are in here because both are true, in different weeks, to different people, and a manual that never contradicts itself has been tidied into uselessness. When two entries pull against each other I say so and I name the conditions. Deciding between them is yours. That is not me avoiding the question. It is the question.

Almost nobody in this book was a fool. The genre before all this was full of collections that assumed otherwise, and they were worthless then

and they would get you killed now. The man who welded the fire doors was a locksmith of thirty years and he was reasoning correctly about intrusion. The group that shot first had buried four people who had not. If your reaction to an entry is that the person was an idiot, the entry has failed and you should read the second paragraph again.

What you need to know about them, in seven lines. It passes in saliva, into blood, by a bite. Not by air, not by water, not on your hands. They do not eat, which is the fact everything else hangs from: they bite once and walk on, so almost everyone bitten lives long enough to turn, which is why there are so many of them and so few of us. Turning takes between four hours and nine days. They are slow, about four miles an hour, and they never stop, and the difference between those two facts is the whole of chapter three. They hunt by sound, then by movement, then by smell, and they use smell to tell each other apart. They run on the air temperature: below four degrees they are slow, in a hard frost they stand still, and in August they are quick and they rot. The brain stops them, or the spine above the shoulders, or enough broken frame that the legs will not carry them.

Learn the seasons before you learn the weapons. Everyone wants the chapter on killing them. There is no chapter on killing them, because killing them is a thing you do when the rest of the manual has already failed. Winter is the working season. Summer is the season you get through. The single most useful sentence I can give you is that a plan made in February will kill you in July, and half the graves I have dug were dug for people who made one good plan and kept it.

How it is arranged. Three parts, by the problem in front of you. The First Week covers the first hour, what to believe, and moving. The Body and the Building covers doors, noise, and the unglamorous business of water, food and teeth. Everyone Else covers other people, your own group, and the long middle, which is what we are in now and what nobody wrote a rule for, because nobody expected to get here.

Chapter nine is the reason for the other eight. It is about the fourth year. It has the fewest deaths in it and it is the chapter people ask me for. Read it last, or read it first if this week has been bad.

On me. I run the Long Room, which was a subscription library in Beeston and is now a store of paper, seed and about eleven hundred notebooks. I did not write these rules. I collected them, checked them where checking was possible, and wrote the two paragraphs underneath. Where an entry is my own I have signed it. Where I have reconstructed a rule from somebody who could not write it down themselves, the note at the back says so.

The Long Room, Beeston. The fourth winter.

I

THE FIRST WEEK

The First Hour

The first hour has almost nothing in common with the rest of this book.

Everything else in here was worked out over months, by people who had the time to be wrong twice and correct it. The first hour was not worked out. It was survived. By people who happened to be in the right building, or wearing boots, or who had gone out for milk. I want that on the first page. A manual that pretends the first hour was won by preparation is lying to you, and the lie costs something. It makes the survivors sound clever. It makes the dead sound careless. It is the reverse of what the notebooks show.

Here is what the first hour was. Between nine and half past eleven on a Tuesday morning in the third week of March, most of this county worked out at about the same time that something was wrong, and then spent forty minutes finding out what, and then made one decision each. That decision is the entire reason some of them are named in this index and the rest are not.

The rules that follow have one thing in common. They are about not spending time. Not spending it well. Not spending it. Every one of them is a way of getting a decision made before you have enough information to make it properly. There was no version of that morning in which you got enough information. The people who waited for it are not here.

Two warnings. The first is that these rules aged badly. Six of the eleven were dangerous by June and I have said so underneath each. The first hour

is the only chapter in this book with an expiry date on it. If you are reading this in the fourth winter, it is history and not instruction.

The second is that the last rule in the chapter is not a survival rule at all. It is in here anyway.

Read them fast. That is the register they were written in.

001

“Leave in the first hour, or make a plan to leave in the third week.”

Priya Raghunathan, A&E nurse, day 1

WHY IT WORKS

Everyone leaves at once. That is the whole of it. A road carries a certain number of cars an hour and no more. The southbound carriageway held for about ninety minutes on the first morning. Then it stopped being a road and became a long thin car park with families sitting in it. It never moved again. The first hour is empty because nobody has decided yet. The third week is empty because the cars have been left where they died and walked away from. In between is the worst place in England. Raghunathan walked out of her department at twenty past nine, against every instinct she had been trained into. She was past the ring road before the first ambulance failed to arrive.

HOW IT GOES WRONG

The first hour is empty because nobody knows anything, and that includes you. She drove north for forty minutes towards a town the news had not reached, which was already gone, and she had to turn round. She lost the hour and she lived through losing it. That is the only reason this rule is at the front of a book instead of cut into a stone. Anybody moving in the first hour is moving on rumour at fifty miles an hour. The third week is slower, colder, hungrier and far more honest. If you have any way of holding out for it, hold out for it.

002

“Whatever you are locking out is in the building with you.”

Marcus Okon, police sergeant, day 1

WHY IT WORKS

Okon cleared eleven premises in two days. In nine of them the doors had been made fast from the inside. By people who had shut the problem in with themselves and then gone to sleep. Turning takes hours. Somebody bitten on Tuesday afternoon is a colleague at nine, quiet at midnight and a catastrophe at four. The lock is on the wrong side of the question. Before you seal a room, count the people in it. Account for every one of them out loud. Look at the arms of anybody who says they are fine.

HOW IT GOES WRONG

It turns you into a man who checks the arms of strangers, and that is a difficult thing to stop being. Okon searched a family of four at the Hunslet checkpoint on day nine, found nothing, and waved them through. He searched them again on day eleven, because by then he could not not do it. The father broke his cheekbone and took the checkpoint's food. I do not blame him. Suspicion is a tool with a very short handle. This rule is for a room you are about to seal. It has no business anywhere else.

003

“Fill the bath. Then fill everything else. Then sit down and think.”

Bea Winslow, retired postmistress, day 1

WHY IT WORKS

The towers are fed by gravity, so water keeps arriving for hours after the pumps have stopped, at a pressure that falls all afternoon. Then one time you turn the tap and it coughs. A bath is eighty litres. A cistern is nine. An older house has a tank in the loft with another two hundred in it that most people never think about. The order in the rule is the clever part: filling things needs

no attention, so you can do your thinking while the taps run. Winslow filled the bath, four buckets, the kettle and two wheelie bins she dragged in over the kitchen lino. Only then did she get out a pencil.

HOW IT GOES WRONG

Eighty litres weighs eighty kilos and you cannot carry a bath. There is a particular death that comes from having done something sensible on the first morning. It looks like a person standing in a well provisioned kitchen in June, unable to think of a reason to walk out of it. Winslow left hers on day thirty-one with a rucksack and a hand cart, and abandoned every drop. Fill the bath. Then do not let the bath make your decisions.

004

“What is on your feet when it starts is what you have got.”

Ewan Tait, mountain rescue, day 2

WHY IT WORKS

You are not going home. That is the part people took three weeks to believe. Tait believed it on the second morning. He had spent nineteen years walking off hills with people who had made exactly one bad footwear decision that day. Feet are the first thing to fail and the hardest to repair. He kept a tally through the first spring of the people he found who could not go on. It comes out at eleven in walking boots and sixty-four in something else. Look down. That is your whole vehicle now.

HOW IT GOES WRONG

It became a superstition. Men who had got out in good boots would not take them off. By the second month there was rot in the county that had nothing to do with anybody being bitten. Boots come off at night. The other failure is Tait's own. Good boots made him confident. Confidence took him over the tops in April with weather coming. Mountain rescue is a thing that requires there to be a mountain rescue.

005

“Fifteen minutes of packing, then out, whatever is still on the table.”

Tomas Vrba, lorry driver, day 1

WHY IT WORKS

Packing expands to fill whatever time you give it, and the time you have is never what you think. Vrba loaded his car in fifteen minutes. He had loaded vehicles for a living for twenty-two years and knew the thing nobody else knows. It is not the carrying that takes the time. It is the choosing. A timer removes the choosing. He set the alarm on his phone, filled the boot, and drove off at eleven minutes past with the tailgate still up.

HOW IT GOES WRONG

Fifteen minutes gets you the obvious, and the obvious is tins. Go through the notebooks and the same three things are left on the table in house after house: documents, medicine, and somebody's glasses. You can eat another man's tins. You cannot borrow his eyes. Vrba got out with four hundred tins and without his daughter's inhaler. He went back for it on day four, which is a thing this book tells you twice never to do. He came back with the inhaler. That is one of two occasions in eleven hundred notebooks where going back worked. It is in here so that you know how rare it is. Not so that you copy it.

006

“Down. Always down.”

Oz Prentice, scaffolder, day 1

WHY IT WORKS

Height feels like safety and is not, because height is only safety if you can get off it whenever you choose. The lifts went in the first week. That leaves one staircase, and they climb stairs, badly and slowly and without ever getting bored. A tower block is a chimney with a single exit and four hundred people in it. Prentice had spent his working life at altitude on somebody else's

building. He had a rigger's instinct about the way down being the part you plan first. He went down on the first afternoon, walked out of Little London, and is alive.

HOW IT GOES WRONG

Down is where the water goes, and the smoke, and the crowd. Down is the street, and the street in the first week was the drift. The eleventh floor of Prentice's own block held out until the first week of May, on rainwater and a tank. Four of those people are alive too. This is the first place in the book where two good rules point opposite ways. Height traps you slowly and the ground kills you quickly. Which of those you would rather have is a real question with no general answer. Anybody who gives you one has not read enough notebooks.

007

"Nobody who is more than one street away."

Rob Hatherleigh, logistics corporal, day 1

WHY IT WORKS

It is arithmetic and it is not pleasant. Every retrieval costs you the journey out, the search, the argument on the doorstep and the journey back. Each of those is exposure. Exposure in the first week compounds. Hatherleigh had spent eleven years moving materiel. He wrote the rule at half past ten on the first morning, before he had seen one of them, on the back of a delivery note. It is the most hated sentence in this book. It is also, in the individual case, correct, and I have never found a set of numbers that says otherwise.

HOW IT GOES WRONG

He went back for nobody. That included his brother in Armley, four streets off, and he was right, and it has eaten him alive for four years. Then there is the group problem. People will not follow a man they believe would not come for them. Hatherleigh's second group lost eleven of nineteen in the autumn. Not to anything that bit them. To nine people walking off in the night because of a rule pinned to a wall. A policy that is correct for one person can be fatal for twenty. Chapter eight is about that and about nothing else.

008

“Their blood will not kill you. Their spit will.”

Yusuf Errai, veterinary surgeon, day 3

WHY IT WORKS

This is the single most useful sentence in the manual and the county had it backwards for the best part of a year. It passes in saliva, into blood, by a bite. Errai worked it out on the third day, from the pattern of who turned and who did not. Everybody bitten, and nobody else. That included two men who had spent a morning up to their elbows taking one apart. Blood on unbroken skin is nothing. Blood on your coat is nothing. Knowing that is the difference between two groups. One can clear a doorway. The other burns its own clothes and starves in a clean house.

HOW IT GOES WRONG

It is true and it has a hole in it. Blood in an open cut, blood in an eye, blood in a mouth. Errai lost his assistant. She was splashed across the face cutting one down from a fence, and she had a torn cuticle on the hand she wiped herself with. There is no version of that sentence that would have saved her. Say the longer one. Their blood will not kill you through skin that is whole. You will not know which of your skin is whole until you check. So check.

009

“Off at the meter, and do the neighbours while you are out.”

Sam Threlfall, plumber, day 2

WHY IT WORKS

Fire took more of this city than they did. Nobody argues about that any more. A hob left on. A pilot light. A gas main under a house nobody is watching, and no brigade. A terrace burns as one long building, with party walls that were never meant to be the last line. Threlfall walked the length of Cross Flatts with a meter key on the second morning and shut off forty-one

houses. Including his own. Including the four he had to break into. It took him ninety minutes and it is the reason that street is still standing.

HOW IT GOES WRONG

Ninety minutes in the open on day two, alone, in daylight, is a fair description of how most of the dead in this book died. He got away with it. Then there is what he built. A street with no gas is a street with no cooking and no heat. By November people were burning furniture in the front rooms of houses that were beautifully safe from gas. Three of them went up anyway. He solved the fire of March and bought the fire of December, which is a trade, and worth making, and worth knowing you have made.

010

“Nobody is coming.”

the Beeston wall

WHY IT WORKS

Three words did more work in this county than any three I know. Everything about a person in a disaster is built around waiting. You ration for a fortnight. You signal. You stay where the rescuers will look. You keep the road clear. All of that assumes an outside that is functioning, and there was not one. Accepting that is the moment a victim becomes a resident. The people who took it in during the first week planted in the first spring. The people who did not were still sitting in the loft in June with a torch and a whistle.

HOW IT GOES WRONG

Somebody came. Twice, that I can document, both in the first month, both by boat. And the sentence turned into a licence, because it never once told anybody what to do next. It was used to justify not answering a knock, not walking to the next village, not looking for a woman’s children. It is one of the four in this book that arrived already famous. No notebook and no body behind it. Worn smooth by three hundred retellings. There is a note at the back about how much that smoothness ought to worry you. It is the best sentence in the manual. That is what is wrong with it.

011

“Some people are staying. Let them, and stay with them if that is yours to do.”

Joyce Trent, hospice nurse, day 2

WHY IT WORKS

Not everybody can be moved. There is a ward of that in every town and there was one on Beckett Street. Nine people who could not walk. Two on oxygen. One who was four days from the end of a long illness that had nothing to do with any of this. Every rule in this chapter says leave. Trent’s answer is that the arithmetic is correct about survival and has no jurisdiction over anything else. She stayed. She kept nine people comfortable for eleven days. In the circumstances of that fortnight that is a piece of work I cannot get to the bottom of. She wrote this on the ninth day, in a drug chart.

HOW IT GOES WRONG

It is a rule that can be used on you. Up and down the county in that first fortnight, people were told they were staying, in a kind voice, by people who were leaving. The sentence in this entry is exactly the sentence they were told. Two of Trent’s own nurses stayed because she had. They were twenty-four and twenty-six and they could have walked out on the first morning. A choice that is noble in the first person is coercion in the second, every time, with no exceptions I have found. So read it in the first person or do not read it at all. Nobody gets to make this one for anybody else. If you are reading it because somebody handed you the book open at this page, go.

What You Believe

More people in this index were killed by something they had been told than by anything that ever laid a hand on them.

That is not a figure of speech and it is not quite a statistic either, because I cannot audit a cause of death from a notebook, but I have counted what I can and it is not close. The bad information came in three kinds and they are worth separating, because they need three different defences. There was the rumour that moved people, which is the deadliest: a camp at Kendal, a working hospital at Harrogate, a ferry. There was the rumour about the biology, which killed slower and in larger numbers: that it was airborne, that fire frightened them, that running water stopped them, that a fever meant you were turning. And there was the rumour about other people, which is chapter seven's problem and which killed the fewest bodies and the most communities.

The thing that made all three so effective was that the ordinary machinery for checking a claim went in the first fortnight, and nobody noticed it going. For a few weeks the radio was still talking and it sounded like the machinery. It was not. It was three men in Skipton with a generator, reading out things that people had shouted at them. They were doing their honest best. They were wrong about the Kendal camp for eleven days, and about four hundred people walked north on the strength of it.

So the rules in this chapter are all, underneath, the same rule in different clothes. Find out who saw it. Everything else here is a technique

for getting to that question faster, or for surviving the fact that you often cannot get to it at all.

One warning before you start. This is the chapter where the entries argue with each other hardest, and I have not smoothed that out. Entry 20 tells you to discount anything you cannot source. Entry 21 tells you that the best trick anybody found was discovered by nobody and has never been sourced by anyone. Both of those are in the book because both of them are true, and living inside that contradiction is most of what thinking is for now.

012

“Write the date before you write anything else.”

Nell Okafor, librarian, day 5

WHY IT WORKS

Time stops behaving. Without a written date the weeks fold up, and the folding costs you three things that matter. You lose the count on a bite, which is the difference between watching somebody and burying them. You lose the season, and this is a book about seasons. And you lose the ability to put two accounts side by side and see which came first. Okafor started the Long Room ledger on the fifth day with the date and the weather and nothing else, because that is what a librarian does when the world ends. Everything in this book has a day count on it because of her.

HOW IT GOES WRONG

She kept the record and did not keep herself. There is no rule in her hand about food and she was seven stone in the second November. Beyond that, dates become a fetish. A group at Otley ran a calendar that had drifted nine days by the second spring. They planted on a written date instead of on a hedge coming into leaf, and lost most of a crop to a frost that the ledger said was over. The date is a tool for comparing accounts. The hedge is the authority on the hedge.

013

“A rumour that tells you where to go is a rumour somebody wants you to be.”

Idris Pyle, day 61

WHY IT WORKS

Ask what a piece of information does when it enters a hundred heads. Most true things do nothing much. But a rumour that moves people moves them all in one direction, at one time, along one road, carrying everything they own, and there has never been a shortage of men who find that arrangement convenient. Pyle knew because he had spent his working life in the trade. Three of the four big movements in this county in the first year had somebody standing at the far end of them. The Kendal camp is the one everybody knows. There was no camp. There had never been a camp.

HOW IT GOES WRONG

He was right and he used it. Two of the four movements were started by Pyle. Then there is the cost to everyone who believed the rule too well. A family on Dewsbury Road turned down three true offers of shelter in one winter, because each of them told them where to go. Two of the four died a hundred yards from a house with a fire in it. Suspicion of good news is expensive and it is not free of charge just because it is clever. Ask who benefits, then go and look at the place from a distance, then decide. Do not stop at the first step and call it wisdom.

014

“Believe the weather. Nothing else.”

Jonah Kite, radio operator, day 12

WHY IT WORKS

It is a rule about verifiability, and it has the merit of being usable by somebody with no training and no time. You can check a forecast by walking outside. You cannot check a report about Harrogate without going to Harrogate. Kite

ran a set on car batteries from a bungalow at Guiseley and listened to everything that came over for eleven months. This is what he had to show for it. The weather was right about four times in five. Nothing else on the air was better than a coin.

HOW IT GOES WRONG

There was one broadcast in the first year that was worth acting on, and it was not the weather. On the ninth of April a woman read out, four times, that the Aire valley was going to be opened up by a dam release, and Kite did not act, because it was not the weather. He lost his set and his bungalow and one of his two neighbours. A rule for filtering noise will filter the signal too, and it will do it on the one day of the year when that matters. Keep the rule and add a second sentence: anything that repeats itself four times gets a look out of the window.

015

“Nothing you hear about them is true unless you can name the person who watched it happen.”

Yusuf Errai, veterinary surgeon, day 30

WHY IT WORKS

Look at what this rule killed. That it travels on the air. That fire frightens them, which cost a barn and nine people at Collingham. That running water stops them, which is a folk belief about something else entirely and which put two men in the Wharfe in February. That a fever means you are turning, which meant that in the first winter people were being put out of doors for having flu. Not one of those had a witness attached to it. Every one had a confident man attached to it. Errai’s rule is the cheapest test there is, and it takes one question, and the question is who saw that.

HOW IT GOES WRONG

The best thing anybody discovered has no author. Nobody knows who first worked out that they leave you alone if you smell enough like them. There is no witness, no notebook, no name. It works. It has saved more people than any other single fact in this manual. Errai’s rule would have thrown it away.

The rule is right about claims and wrong about techniques, and the difference is that a technique can be tested behind a wall by one person in an afternoon. Test what you can test. Doubt what you cannot.

016

“Two sources or it does not go on the board.”

Rob Hatherleigh, logistics corporal, day 44

WHY IT WORKS

He ran a board in the Wetherby depot with three columns: heard, confirmed, acted on. Nothing crossed from the first column to the second without a second mouth that had not spoken to the first. It is the discipline of an operations room and it is the reason his group never once walked into an empty camp, in a year when four hundred people did. When a group is deciding whether to move eleven miles on a stranger's word, a board on a wall is worth more than any amount of good judgement in one man's head.

HOW IT GOES WRONG

Two people repeating one rumour is one source wearing a coat, and telling the difference takes exactly the skill the board was built to replace. Worse is what it does to speed. On day 202 a boy came in and said the Aire had come over at Kirkstall, and there was nobody to confirm him. It sat in the first column for two days. By then eleven people were in the water. Hatherleigh added a line to the board that night, in different ink, and it is still there: a warning goes up on one source and comes down on two.

017

“Ask them what they were doing when they saw it.”

Marcus Okon, police sergeant, day 18

WHY IT WORKS

It is the oldest question in the trade and it works because of how memory is built. A person who saw a thing was in the middle of something else at the

time: carrying a box, arguing, looking for a dog. That detail comes back with the event attached to it and it comes back cheap and specific. A person who is repeating a story has no such detail, because there was no such afternoon. Okon separated true accounts from passed ones at the Hunslet gate for eight months, and it took him one question and about ten seconds.

HOW IT GOES WRONG

Anybody good enough to be worth catching has thought about this. Pyle's accounts are full of the dog and the box, and they are full of them because he understood what the question is for. And the test punishes a particular kind of honest person: the shocked one, who has no detail at all, and whose account is flat and thin because it happened three hours ago to them and they have not slept. Twice in this book somebody was turned away at a gate for telling the truth badly. Use the question to find people to trust. Do not use it to find people to refuse.

018

“The map is four years old. So are you.”

Callum Reith, day 1,340

WHY IT WORKS

Reith is the only person in this book who has spent more of his life in this than before it, and it makes him worth listening to on a subject none of the rest of us are honest about. A map is a claim about a place, made at a date, by somebody who is not there now. The Kirkstall terraces are a field. Two viaducts are down. There is a lake where the retail park was, and it is on no map and it is nine feet deep. He walks with a map and treats it as a rumour with contours.

HOW IT GOES WRONG

He went the other way with it and stopped believing written things at all, which is a young man's error and an expensive one. In the third summer he walked into the works at Knostrop, which is marked, clearly, on the four year old map, as a sewage treatment plant. He came out with something in a cut on his shin that nearly took the leg, and he was carried nine miles by two

people who could ill afford the day. A map is out of date about buildings. It is not out of date about the ground, or the water, or where the poison was.

019

“Say that you do not know.”

Priya Raghunathan, A&E nurse, day 96

WHY IT WORKS

She came out of a trade where the words are said out loud and where saying them is a mark of competence, and she brought the habit with her into a world running on men who were certain. Here is the mechanism. A group can absorb the news that its leader does not know something. What it cannot absorb is discovering, later, that the leader did not know and said otherwise, because after that nothing that person says can be spent. Uncertainty said out loud is cheap. Uncertainty found out is the end of your authority in one afternoon.

HOW IT GOES WRONG

There is a moment when a group needs a decision more than it needs an accurate account of your confidence. She hit it at Wetherby in the second January, with fourteen people standing in sleet waiting to be told which way. She told them what she did not know, at length, and she was completely honest, and they stood in it for forty minutes and two of them lost fingers. Say you do not know, then say what you are going to do anyway. The second half is not optional and it is the half she left out.

020

“Anybody certain about the biology has never watched one turn.”

Nadia Rahman, pharmacist, day 40

WHY IT WORKS

In the first spring every third man was an expert. The confidence was inverse to the exposure, exactly and reliably, and it is the cleanest example of the effect I have ever seen. The people who had sat with somebody through the six days would not tell you the shape of it, and would say the thing about how it varies, and were hedging because the world was hedging. Rahman had counted out other people's medicines for nineteen years and had a professional's ear for a man who has read one thing.

HOW IT GOES WRONG

It disarms the person who did the work. Errai had the transmission route correct on day three, from evidence, and was dismissed for months by people quoting Rahman's rule at him, because he was certain and certainty was the tell. She has said since that it is the thing she regrets. Doubt a man's certainty, then ask him what it is built on, and if the answer is a pattern he counted himself then he is not the problem. The rule catches the confident. It does not distinguish between the confident and the correct, and you have to do that part yourself.

021

“Keep a book of what you were wrong about.”

the Skipton radio log, day 150

WHY IT WORKS

Three men with a generator broadcast the Kendal camp for eleven days and about four hundred people walked north on it. What they did afterwards is the reason they are in this manual. They kept a second book, read out every Sunday, of everything they had said that had turned out to be false. By the

second year they were the only voice on the air that anybody in three counties trusted. You cannot get better at judging a claim without a record of your own misses, because memory quietly deletes them. Write them down and the deleting stops.

HOW IT GOES WRONG

A list of a man's errors, in his own hand, is a weapon lying in the open. Skipton's book was used in the second winter to break the youngest of the three operators in front of a room of forty people, by somebody who wanted his position, and it worked, and he left. If you keep the book, keep it for one person. A group's book of its own errors is a fine idea and I have never once seen it survive a disagreement about food.

022

“You are allowed to believe something because you need to.”

Sister Agnes Moray, day 300

WHY IT WORKS

Four years is a long time to run on evidence alone and I have not met anybody who managed it. Moray's position, and she is careful about it, is that a belief has two properties and they are separate: whether it is true, and what it does to the person holding it. The man on Tong Road who is sure his daughter reached Ireland gets up in the morning. The one who worked out the odds does not. She is not telling you the odds are wrong. She is telling you that you are a piece of equipment that has to keep working, and that the maintenance of it is a legitimate thing to spend belief on.

HOW IT GOES WRONG

The moment a belief starts costing anybody anything, it has stopped being maintenance and become a plan. Eleven people walked to Holyhead in the second autumn on a belief of this kind and four came back. And the sentence is the exact sentence that a certain sort of man says at the beginning of the thing he builds around himself. There have been two of those in this county and one of them is still going. Moray's own guard against it is worth copying.

She holds one belief this way, she knows which one it is, she can name it, and she has never once asked anybody else to hold it with her.

Moving

Here is the arithmetic that everything in this chapter comes out of, and I would rather you had it in one paragraph on one page than scattered through eleven entries.

You walk at three miles an hour on a road with nothing on your back, and at two and a half with a load, and at less than two over ground. You stop. You sleep. Over a full day and night, hard going, you will make twenty miles and be ruined. They walk at a shade under four. They do not stop, they do not sleep, they do not sit down at four in the afternoon and take a boot off. Over that same day and night one of them will cover something close to ninety miles, if it has a reason to. Put those two numbers next to each other and the conclusion is not comfortable and it is not negotiable. Over any distance that matters, you cannot outwalk them. You never could. Every plan built on getting far enough away has failed.

What saves you is the last four words of that sentence, if it has a reason to. They follow what they can hear, then what they can see move, and they follow it while it is there and they lose it when it is not. A thing that stops being perceptible stops being followed, and within an hour or two the one that was coming for you is drifting again with the rest. So the distance you need is not the distance to safety. It is the much shorter distance to the corner, the hedge, the wall, the far side of the wind. You are not running away. You are becoming uninteresting, and the running is only the part of it that gets you out of sight.

Everything else in this chapter is about feet, weight, water and hours, which are the four things that decide whether you get to the corner. None of it is exciting. Six of the eleven people in this chapter were walkers, drivers or riders before, and none of the ones who were not lasted the first year, and I do not think that is chance.

023

“You cannot outwalk them. You can only stop being worth following.”

Ewan Tait, mountain rescue, day 20

WHY IT WORKS

Tait did the sum on the second Sunday, on the back of a leaflet, and it is the sum at the top of this chapter. Once you accept it, everything about how you move changes and it changes for the better. You stop thinking in miles and start thinking in corners. You stop grinding out an extra four miles at dusk, which is exactly when the ground gets worse and your judgement goes. You spend that energy on breaking line of sight, once, properly, and then walking away quietly. The people still alive in this county move in short deliberate legs between places they cannot be seen from. Nobody is doing forced marches. The forced marches are all in the notebooks that stop.

HOW IT GOES WRONG

It is a rule for one of them, or four. It is not a rule for a drift. A drift of two thousand does not lose you, because the ones at the back are following the ones at the front, and the front is a mile wide. There is no corner that takes you out of sight of all of it. Tait knew this and wrote it, and the sentence that got copied around three counties is the short one, without the qualifier. I have had it quoted at me by a man who was proposing to walk across the front of a drift on the strength of it. Against a crowd, distance is the only thing that works, and you had better start early and you had better have somewhere to be.

024

“Twelve miles a day with a load, and no arguing about it.”

Rob Hatherleigh, logistics corporal, day 12

WHY IT WORKS

It is a real number and it comes from a century of people measuring what a column can do day after day without falling apart. Twelve miles with weight is sustainable. Fifteen is a good day. Twenty is a day you pay for over the following two, and armies worked this out long before any of us needed it. The value of writing it down is that it stops the argument at four in the afternoon, when somebody is feeling strong and wants to push to the next town. The person feeling strong is never the person who will fail. Hatherleigh's group made the same distance every day for five weeks and arrived able to work.

HOW IT GOES WRONG

It is a soldier's number, taken from fit men of similar age carrying similar loads, and his group had a woman of sixty-three, a boy of nine and a man who had been a bank clerk. A column moves at the speed of the person who is struggling, and if the number on the wall says twelve then what happens is not that everybody does twelve. What happens is that the woman of sixty-three does twelve and is destroyed by it, in silence, for four days, because she can read the number too. Set the day's distance from the slowest pair of legs and set it out loud. Then write that number on the wall.

025

“Never the road. The thing beside the road.”

Fen Doherty, poacher, day 30

WHY IT WORKS

A road is the best navigation there is and it costs you everything else, because it is a corridor. Every walker, every drift and every man with a rifle and an idea is using the same corridor for the same reason. What Doherty worked out in

his first fortnight is that you can have the navigation without the corridor. Walk the hedge line, the ditch, the field margin, the strip of scrub inside the fence. You are forty yards off. You can see the road and it cannot see you. You keep the direction and you keep the cover, and if something comes down the tarmac you lie down in a ditch and it goes past. He crossed the county four times in the first year and was seen twice.

HOW IT GOES WRONG

It is half the speed on a good day and a third of it on a wet one, and a hedge line in February is a serious piece of work. The margins are also where everything is: the wire, the old wells, the drainage cuts under the grass, the badger holes. Doherty went into a rabbit hole on the flat, in daylight, in the second spring, and put his ankle out nine miles from anybody. He says himself that a man on the road would have been at Tadcaster by four. The road when you are strong and the time is short and you can see a long way. The hedge the rest of the time.

026

“Follow the cut.”

Bill Cattermole, lock keeper, day 25

WHY IT WORKS

A canal is the best walking in England and it always was. It is flat, because it has to be. It is waymarked to the quarter mile. It goes between towns without going through them. There is water in it, there are bridges every half mile that let you swap sides, and there are gates and locks that make a natural place to stop and see what is behind you. Cattermole had worked the Leeds and Liverpool for thirty-one years and could walk it in the dark, and eleven people came down it with him in the second April and every one arrived.

HOW IT GOES WRONG

A towpath is a bottle. On one side is a hedge or a wall and on the other is the cut, and if something comes at you from both ends of a straight you have two options and one of them is the water. Here is the part people learn once. They do not drown. They do not float, they do not panic and they do not need to

breathe. What happens when one goes in the canal is that it walks along the bottom. It comes out on the far bank six or seven minutes later, and it comes out where you are. Two of Cattermole's eleven found this out in the same afternoon. The cut is the best road in the county and it is the worst place to be surprised.

027

“Boots off at every stop. Every stop.”

Priya Raghunathan, A&E nurse, day 33

WHY IT WORKS

Skin that is wet for long enough stops being skin. It goes white, it goes soft, it comes off in sheets with the sock, and then you are a person who cannot walk in a world where walking is the only thing there is. Raghunathan saw four of these in one week in the first April and wrote the rule that night. Off, out, wrung, feet dry in the air for ten minutes, dry pair on, wet pair inside the coat against the body where they will be half dry in two hours. Ten minutes an hour is expensive. A foot is more expensive, and unlike almost everything else in this book, a ruined foot cannot be traded for, prayed over or fixed later.

HOW IT GOES WRONG

There is a stop where you cannot afford eight minutes and you will not know it is that stop until it is. A man called Devlin died at Micklefield in the second November sitting on a wall with both boots off and both socks in his hands, and he had done everything right for two hundred days. The rule needs its exception said out loud so that people can use it: boots off at every stop where somebody else is watching the ground. Alone, or with one other, you dry your feet at the place you are sleeping and nowhere else.

028

“Move at four in the morning.”

Kit Anand, courier, day 15

WHY IT WORKS

Four in the morning is the coldest hour of the day, and cold is the one thing that reliably takes them apart. Below four degrees they are slow and stupid. In a hard frost they stop where they stand and become street furniture that you walk around. So the coldest hour is the hour when a thing that is faster than you is not faster than you. It is also the hour when every human being for nine miles is asleep, which in this county matters at least as much. Anand had been riding a bike round this city at that hour for six years before any of it and knew the streets in the dark better than in the light.

HOW IT GOES WRONG

You cannot see, and everything in the last four entries about the ditch and the wire and the ankle is four times as likely in the dark. The cold that slows them slows you first and slows you more. You are the one who has been asleep and is now walking at two miles an hour with no breakfast. Hypothermia does not need a hard winter to work on a thin person at four in the morning. And in July four in the morning is fourteen degrees, and they are as quick then as at noon. This is a rule for October to March. In summer it is just walking in the dark for no advantage, and Anand herself stopped doing it in the second June.

029

“Carry the means to drink, not the drink.”

Gwen Prydderch, water engineer, day 55

WHY IT WORKS

Water is two and a bit pounds a litre and you need three or four litres a day, and if you carry that you are carrying eight or nine pounds of the twenty-five you can carry at all. But this is England. There is water every mile, in gutters,

streams, tanks, roofs, radiators and the header tank of every loft you pass. What you lack is not water, it is water you dare drink. Prydderch carried a litre, a filter, a bottle of tablets and a small pan, which came to under three pounds, and she drank six litres a day for two years and was never once ill. The weight you save goes into food, which is the thing there is genuinely not enough of.

HOW IT GOES WRONG

Means fail and water does not. A filter clogs, and it clogs without warning on the day you push it through silt. Tablets get wet. A pan needs fire and fire is chapter five's problem. In the second August there was a stretch across the flat land east of here with twenty miles and nothing in it, in a summer when everything standing had gone green and foul. The man carrying four litres of stupid heavy water walked it. The two people carrying filters did not. Carry the means. Then carry a litre anyway, and be aware of what it is for, which is the day the means let you down.

030

“Walk to somewhere. Never away from something.”

Lila Semenova, climber, day 8

WHY IT WORKS

Retreat has no end condition. That is the whole argument and it is a good one. A group that is moving away from a thing will keep moving, because there is no distance at which anybody can say we have arrived. So it walks until it is too tired to make decisions, and then it makes one. A named place fixes that. It fixes the day's distance. It fixes the direction when you are tired. It gives you something to talk about at night that is not what you left. And it lets the argument happen at the map, and not at four in the afternoon in the rain. It does not have to be a good place. It has to be a place with a name.

HOW IT GOES WRONG

A named place is a thing you press on towards, and everybody who has climbed knows what that does, and Semenova knew it better than anybody in this book and did it anyway. Her group had Malham in their heads. They kept

the number of days it would take and they kept it after the weather turned and after the older of the two brothers started coughing. Turning back means admitting the plan was the problem, and the plan was hers. They got to Malham. Two of the five got to Malham. A destination is a tool for making decisions and it becomes, without anybody noticing the hour it happens, a reason not to make them.

031

“Nothing in the bag that you have not carried up a hill.”

Pearl Adjei, bus driver, day 6

WHY IT WORKS

Everything is light for the first mile. That is the trap. It is why every abandoned pack on every verge in this county has the same four things in it, and why the verges of the first spring were a museum of what people thought they needed on a Tuesday. Adjei loaded her bag on the second day, walked it up the hill at Beeston and back down, and took out a third of it. She did that three times over a fortnight, until what was left was what she could carry all day. Twenty-five pounds is a load. Thirty-five is a load you will put down in a hurry one day, and the day you put it down is the day you needed it.

HOW IT GOES WRONG

Watch what people cut. They cut the thing they have not used yet. The things you have not used yet are the medical kit, the spare socks and the second knife, and those are the items whose entire purpose is the day that has not happened. Adjei cut her kit down to a beautiful weight over three weeks and had no dressings in it at all when a woman in her group opened her hand on a gate in June. Weigh the bag. Then take the medical kit out, put it on the floor, and pack it first, and let everything else compete for what is left.

032

“Run for the corner. Then walk.”

Callum Reith, day 900

WHY IT WORKS

It is the practical form of the sum at the top of this chapter. Running does not get you away, because nothing gets you away. Running gets you round the corner, over the wall, into the alley, out of sight, and then its whole job is done. Continuing to run is now working against you, because running is loud, and loud recruits. The thing you have just lost has friends within earshot who have not seen you at all yet. Sprint for as long as it takes to break the line. Then stop, breathe through your nose, and walk out of the area at a speed that makes no noise on gravel. Reith has done this several hundred times and he is twenty years old and he is very hard to kill.

HOW IT GOES WRONG

The corner has to be a real corner. A corner that leads into a yard is not a corner. It is a room with a door in it. The whole method depends on knowing what is round the thing you are about to run round, which means it belongs to people who know the ground. Reith grew up on this ground. The woman he was with in the second winter did not, and she could not do the second half of it. She kept running because everything in her body was telling her to. The noise of it across two hundred yards of car park brought in what was in the multi storey. He got round the corner and walked. She did not. He has written about it once and I am not going to add anything to it.

033

“Stop walking.”

Maggie Lunt, hill farmer, day 400

WHY IT WORKS

The roaming years killed more people than they saved and the notebooks are not subtle about it. Walking is exposure, every day, all day, and it produces

nothing: no store, no crop, no roof, no fence, no relationships with anybody who might help you in February. Lunt watched three groups pass her gate in the first year, all of them going somewhere better. She was still on her land when the third of them came back past, going the other way, with fewer people in it. Everybody alive round here at scale is somebody who stopped. The rule gives you permission to be the person who is not moving, in a year when moving looked like the only competence there was.

HOW IT GOES WRONG

She stopped on eleven acres with a spring on it, a stone barn, walls she could see over and a slope that meant nothing could come at her out of dead ground. That is not stopping. That is stopping somewhere, and the somewhere is doing all the work in the sentence. Two settlements I could name stopped in the wrong place and stayed out of the same instinct. They defended it past the point where anybody with clear eyes would have walked out, and were finished in the second summer inside a week. Stopping is right. Being unable to leave the place you stopped is the same disease as never stopping at all, and it is chapter four's subject, and Lunt keeps a bag packed by her door to this day.

II

THE BODY AND THE BUILDING



Doors, and Where You Sleep

Almost everything people built in the first year was built against the wrong thing.

The instinct is to build against them, and the instinct is understandable, and it is a poor use of the two weeks of work you have in you. They are not a siege engine. One of them at a door is a nuisance that a wedge and some patience will see off. A hundred of them at a door is a situation that no wall a family can build in a fortnight was ever going to hold. So the entire middle ground that people spent their labour on turns out to be the part that never comes up. What actually comes through a wall is fire, water, cold, and people, in that order by frequency and in a different order by how much it hurts.

Here is the correction, and every rule in this chapter is a version of it. A shelter is not a fortress. It is a machine with four jobs. It keeps the weather off you. It keeps you warm on the fuel you have. It tells you something is coming before the thing arrives. And it lets you leave. That last job is the one everybody deletes. It is deleted because deleting it feels like commitment and looks like competence. It is the single commonest fatal design error in these eleven hundred notebooks. I have put it at the front of the chapter and again at the back, in the hope that saying it twice does something.

There is a second correction and it is quieter. The winter kills more than the door does. In the first February, in this county, more people in these papers died of the cold in a building than died of anything with teeth, and most of them died in large rooms in fine houses. They were

heating a volume of air that nothing they owned could ever have heated. The rule about the small room is the least dramatic entry in this chapter and it is the one that has saved the most people. You will not want to follow it in November. You will be glad of it in March.

034

“Count the doors before you count the exits.”

Terrance Bello, locksmith, day 9

WHY IT WORKS

Bello's point is that people survey a building for how things get in, which is a list of doors, and then believe they have surveyed it. The useful list is a different list with different members. An exit is a door you can go through, in the dark, in a hurry, carrying somebody, with whatever tool is in your pocket. A door you have screwed, barred, filled or bricked is not on that list any more. He walked buildings with two counts in two columns and made people say both numbers out loud. In a leisure centre, a chapel and two schools those numbers came out very far apart.

HOW IT GOES WRONG

He counted eleven doors in the leisure centre at Cottingley and made all eleven fast. When the fire took the east stand there were eleven doors and no exits, because he had welded the difference out of the building. That is the whole of it, and he was the best man at that trade in this half of the city, and he had been reasoning correctly the entire way. What he lacked was a rule about which of the two numbers is allowed to reach zero. Make the doors hard. Leave two of them, on different walls, that a frightened person can open with their hands, and tell everybody in the building which two they are.

035

“Wedge. Do not lock.”

Marcus Okon, police sergeant, day 55

WHY IT WORKS

A wedge under a door that opens towards you is close to unbeatable by anything without a shoulder and a plan, and it has four properties a lock does not. It is silent going in. It needs no key, so nobody is ever the wrong side of it. It works on any door in any building you walk into, including the one you have been in for eleven minutes. And it comes out with a kick when you have decided to go, which no bar, screw or barricade does. Okon carried six of them cut from a fence post and got through the whole of the second year on nothing else.

HOW IT GOES WRONG

Wedges want a floor. On carpet they ride. On a worn stone flag they skate. On a door with a quarter inch under it they do nothing at all, and you find this out at two in the morning. They are also a fine defence against a push and a poor one against a sustained lean. That is what a crowd is. A crowd does not push. It arrives and continues to be there. Okon's own note is that a wedge buys you between forty seconds and four minutes, and that the number is not the point. The point is that the forty seconds are yours to use, and if you have not decided in advance what you are going to do with them then you have bought nothing.

036

“Glass is not a wall. Glass is a curtain.”

Oz Prentice, scaffolder, day 22

WHY IT WORKS

Go and look at a house that was lost in the first year and the door is nearly always intact. People fortify what they think of as the way in. The way in was the bay window in the front room, four feet of nine hundred millimetre float

glass that a wheelie bin goes through by falling on it. Prentice had spent twenty years working outside buildings and had a glazier's view of what a house actually is, which is a light frame with big holes in it and some plaster. His rule reorders the work: the ground floor openings first, all of them, before anybody touches a door.

HOW IT GOES WRONG

He boarded a whole ground floor in Armley in three days and then had to live in it. A house with every opening blinded is a house where nobody can see out. Not being able to see out is worse than most of what you were afraid of seeing. Every decision about going outside is now made blind. The people in that house began to go out to find out what was happening, which is the exact behaviour the boards were meant to prevent. Board the glass. Then cut a slot at eye height in two boards on two different walls, and accept the small risk that comes with them. The alternative is a group that has to open the front door to learn anything.

037

“Sleep where you have to be lifted to.”

Lila Semenova, climber, day 24

WHY IT WORKS

They do not climb. They manage stairs, badly, because stairs are a floor at an angle and their legs still work. A ladder is not a floor at an angle. Anything that needs a hand on a rung and a decision about where the next foot goes is outside what they can do. It always has been. It is the most reliable single fact about them after the cold. So: a hayloft with the ladder pulled up. A flat roof reached by a scaffold plank you draw in. A first floor with the stairs taken out and a rope ladder in the hole. Any of those turns eight hours of unconsciousness from the most dangerous part of your day into the safest.

HOW IT GOES WRONG

People. A rope ladder is a wonderful defence against a thing that cannot think, and a signpost to a thing that can. Semenova was robbed twice in a loft she could not get out of, by men who stood at the bottom and waited. It took

them no effort at all. There is also the ordinary failure. She broke a wrist coming down in the dark at speed in the second November, in a hurry, for a reason that turned out to be a fox. Every night for a year she had climbed up and been safe. It took one descent to cost her four months of work.

038

“Two ways out of any room you sleep in.”

Rosa Kaminski, schoolteacher, day 70

WHY IT WORKS

She ran fire drills for twenty-two years and she brought the finding with her. What people do in the first eleven seconds of being woken by an emergency is go back the way they came in, all of them, together. If that way is blocked they stop and stand in it. A second exit does not merely double your chances. It halves the crowd at the first one, which is the thing that kills. Kaminski made every group she was ever part of walk both exits, in the dark, twice, on the night they arrived, and no group of hers has ever lost anybody in a building.

HOW IT GOES WRONG

Two ways out is two ways in, and the whole of chapter four is an argument about that trade. Her old school had fourteen exits and was the most indefensible building in the district, and she is clear that she would not sleep in it. There is a second failure that is subtler. A group that has drilled its exits will use them. Using an exit means going outside at night with no information. Twice in these notebooks a group has gone out of the back door into something worse, because a bin fell over at the front. The drill is right. Add one word to it: everybody goes to the second exit, and nobody goes through it until somebody has looked.

039

“The smallest room you can stand.”

Solomon Quaye, barber, day 200

WHY IT WORKS

This is the entry that has saved the most lives in this chapter and it will not feel like it in November. A body puts out about the heat of a bulb. Three bodies and one candle will hold a box room above freezing all night with the door shut. Nothing you own will do that for a Victorian front room with a nine foot ceiling and a chimney in it. In the first February people froze in beautiful houses with fuel still in the yard. They were heating a volume instead of a space. Quaye moved five people into a back scullery eight feet by seven and hung blankets over the door and the window. They came through the worst month any of us have seen without a single case of anything.

HOW IT GOES WRONG

Five people in a room eight feet by seven, in the dark, for four months, is a thing that happens to your mind. Two of his five stopped speaking to each other in that scullery over something neither of them can now reconstruct, and one of them left in the spring and did not come back. Small rooms preserve the body and grind the group, and the grinding does not show up until March, and by March it is done. Take the room. Then build the day around getting people out of it in ones and twos for an hour, in daylight, doing something with their hands, whatever the weather is doing. Treat that hour as fuel and not as leisure.

040

“Bells.”

Isaac Bramall, beekeeper, day 40

WHY IT WORKS

You cannot watch. That is the fact underneath every perimeter anybody has ever built. A group of six cannot keep a pair of eyes awake and useful on three

approaches for six months. The ones that tried burned out in a fortnight and then slept through everything. So the ground watches instead. Bramall ran fine wire at shin height through the hedges at thirty yards out, with tins and two brass bells off a harness. He could sleep. He had not slept properly since March. The distance matters more than the noise: thirty yards is far enough that the sound reaches you before the thing does and near enough that you know it is your problem.

HOW IT GOES WRONG

Alarm fatigue, and it is merciless. Foxes, cats, wind, a badger every night for a week. By the fortieth false alarm nobody is getting up, and the wire is now a thing that ruins everybody's sleep without protecting anything. Bramall's group had stopped rising to it well before the night that mattered. The fix he arrived at in the second year is the one worth copying: two lines, at different distances, with different sounds. The far one is information and you note it and stay in bed. The near one has never once been an animal, and when it goes everybody moves.

041

“Leave the bodies in the street.”

Fen Doherty, poacher, day 88

WHY IT WORKS

It is the most unpleasant rule in this book and it is sound. A street with bodies in it reads, at a hundred yards, as a street that has already been done. The thing you most want in the second year is to be uninteresting to people. It does more than that. The smell covers yours, which matters, because smell is how they sort each other out. Doherty lived a whole winter on a terrace that four separate groups walked past, and one of them stopped at the end of it, and looked, and went the other way.

HOW IT GOES WRONG

Rats came, and then everything rats bring, and by the second August the flies were a serious medical problem and two of his household had something they could not shake. There is also what it does to people to live in it, which he

does not write about and which the people who were there do. And once, on his own doorstep in the second March, one of them was not a body. It had been lying in the same position for eleven days in the cold, doing nothing, because in the cold that is what they do, and it stood up when the thaw came. Cold ones are not dead ones. If you are going to use this rule, you put a spike through every head on your street. You do it in the frost, when it is safe to. And you write down how many there are.

042

“The smoke, not the flame.”

Duncan Ferrier, gamekeeper, day 60

WHY IT WORKS

Everybody worries about the fire being seen and hardly anybody does the arithmetic on the smoke. At night a flame in a window is visible for two or three miles and you can cover a window. By day a column of smoke off a wet fire is visible for eight or nine, and you can cover nothing. In the first winter you could stand on the ridge above here at ten in the morning. And count the households. Ferrier's method is the one the estates used for a century. Burn dry hardwood. Burn it hot and small. Burn in the middle of the day, when the sky is already grey and the air is turbulent. Put it under a long cold flue that breaks the column up.

HOW IT GOES WRONG

It gives you one hot meal a day, at a time chosen by the weather. It is very hard to run a household of children on a fire that is only lit under cloud. The second failure is the one that got him. His flue trick works because moving air pulls the plume apart. On the still bright morning of the fourth of December there was no moving air at all. The smoke went up out of that chimney in a rope you could see from Otley. He had used the same fire for a hundred days. Weather is not a background condition. It is a member of the household, and it gets a vote every morning.

043

“The best wall is a bad approach.”

Maggie Lunt, hill farmer, day 120

WHY IT WORKS

You cannot build a wall. Be honest about the labour. A family, in a fortnight, with hand tools, cannot build anything that will hold against a serious push. Every hour spent on it is an hour not spent on the store and the crop. What you can do is choose ground where the approach does the work for free. A single track with a drop on one side. A spur with water on two. A slope steep enough that anything coming up it is slow and in the open and visible for four minutes. Lunt's place has one way in, uphill, in view of the house for three hundred yards, and she has never had to defend it because nothing has ever got near it unseen.

HOW IT GOES WRONG

A bad approach is bad in both directions. Everything she owns has come up that track on her back or on a pony. In the second winter the snow filled the cut, and nothing came up or down it for five weeks. There was a woman in that house who needed a doctor. And good ground is legible to anybody who reads ground. Twice in these notebooks a settlement has been found by people with rifles, because the terrain around it was the terrain anybody would choose. That is a strange and specific way to be caught. It has happened more than once.

044

“It is a house. Leave it.”

Bea Winslow, retired postmistress, day 31

WHY IT WORKS

She spent thirty days making a house in Norfolk safe, and it was good work. On the thirty-first she put four things in a bag, walked out of it and never went back. The water had gone bad and she could count. That is the whole

content of the rule and it is the hardest sentence in this chapter to actually perform. A shelter you have worked on becomes a thing you are unwilling to lose, and the unwillingness arrives quietly and dressed as sense. By the time it is obvious, you are a person arguing that the water is probably fine. Everything you built was to keep you alive. When it stops doing that, it is timber.

HOW IT GOES WRONG

Leaving is a real cost and not a free move. There is a kind of person who reads this rule and becomes unable to commit to anything, and drifts for three years accumulating nothing, and is thin. Winslow left one house in four years and settled the second time. The rule is not walk away easily. It is that the walking away must remain possible. The bag stays packed. The two exits stay clear. And the decision gets made on the evidence rather than on the labour already spent. She keeps the key to the Norfolk house on a string. She has never suggested going back for it.

Noise

Everybody learns in the first week that noise brings them, and then spends three years being wrong about which noise.

The wrongness is specific and it is worth an entire chapter, because the mistake runs the same way every time: people are frightened of the sounds that are loud where they are standing, and the sounds that travel are not those. A dropped spanner on a stone floor is startling and it is a sharp, high, short sound, and high sounds are eaten by air, hedges, walls and the ground itself, and forty yards away it is not there at all. A small generator is boring and you stop hearing it in ten minutes. It is low, and low sounds bend round buildings instead of stopping at them. It goes on for six hours. It is audible across a mile of this valley to anything with ears. One of those two sounds has emptied a village. It is not the spanner.

Three facts will do most of your thinking for you. Sound loses six decibels every time the distance doubles, which sounds severe and is much less severe than people expect over the first few hundred yards. High frequencies die faster than low ones, so what reaches a listener at a distance is never the sound you made. It is the bottom half of it. Not the words but the shape of a voice. Not the chop but the thud. And on a clear still night the air a few feet up is warmer than the air at the ground. That bends sound back down instead of letting it escape. The same noise made at two in the morning goes two or three times as far as it did at noon. That last one killed people for a year before anybody wrote it down.

The rest of the chapter is technique. But I want the argument of it stated here, because eight of these eleven entries pull towards silence and

the last two pull hard the other way, and the last two are right. Absolute quiet is not a survival strategy. It is a thing a group can hold for about five months. After that it has produced nothing, built nothing, warned nobody and stopped speaking to itself. I have opened up two houses where everybody inside was thin, mute and still following the rule.

045

“Low and long carries. Sharp and short does not.”

Femi Adeyemi, live sound engineer, day 35

WHY IT WORKS

He had spent eleven years putting sound into rooms and knew, without having to reason about it, which parts of a noise survive a journey. Air absorbs the top end. Foliage absorbs the top end. A brick wall stops the top end and lets the bottom walk round the corner. So the practical ranking is not by how loud a thing is in your hand. It is a generator, a diesel engine, a pump, a big steady voice, a hammer on wood, an axe. And then a very long way down, everything sharp and small that people are afraid of. Adeyemi ranked the household's noises on a sheet of paper in the second month, and the sheet is still pinned up in three settlements I know of.

HOW IT GOES WRONG

Distance is one axis and novelty is the other, and he built his method on the first and forgot the second. What turns a head near you is not the loudest thing, it is the thing that was not there a second ago. A steady hum becomes background inside a minute, to them as much as to you, and a single tap in a silent street at four in the morning is the most attention-getting sound a human being can make. He was killed by neither, as it happens. He was killed on the ridge above Bramley by a piece of corrugated sheet that he stood on, which is short, sharp, and was the only sound in a quarter of a mile.

“Shoes at the door, felt on everything, cloth in the cutlery drawer.”

Bea Winslow, retired postmistress, day 90

WHY IT WORKS

A house makes a sound whether anybody is being careful or not, and it is made of about thirty things: two hinges, a latch, four floorboards, a stair tread, a bucket handle, a kettle lid, the drawer. Winslow's method was to find all thirty in one afternoon and fix them, because a fault you have fixed costs nothing thereafter and a fault you are being careful about costs attention every hour of every day for four years. Grease on the hinges, felt under the bucket, a strip of blanket over the drawer runners, and a nail through the loose tread. After that the household can move normally, which is the entire point and the part people miss.

HOW IT GOES WRONG

A silent house is a house you cannot hear anybody moving in, including somebody who is not supposed to be there. Winslow's second household had a man walk the length of the ground floor and out again with most of a week's food. Four people were awake upstairs. She now leaves one board loose on purpose, at the bottom of the stairs, and everybody knows which. The other cost is speed. People who have been trained out of noise move slowly at exactly the moments when they should be moving fast. There is at least one death in these notebooks that comes down to a woman closing a door quietly instead of running through it.

047

“Loud work in the rain.”

Erin Mackay, fisherwoman, day 50

WHY IT WORKS

Rain on leaves and roofs is broadband noise across the whole range, which is the only thing that masks properly, and hard rain will take twenty feet of chopping and reduce it to something you cannot pick out at a hundred. Wind in trees does the same job less reliably. A river does it all the time in one place, which is why the mills were always the noisiest buildings in England. Mackay ran a rule that any job involving an axe, a hammer, a saw or a scream waited for weather, and she got her boat repaired, her shed built and two winters of firewood done in the rain.

HOW IT GOES WRONG

Masking works both ways and there is no version where it does not. You cannot hear them either, and you cannot hear a person, and hard rain takes your eyes down to about sixty yards as well. She lost a man off the staithe at Paull in the second October to something that walked up behind him in weather that was doing exactly what she had planned for it to do. Rain is also when everything is slippery and cold and your hands are bad. Do the loud work in the rain. Put somebody on the corner who is doing nothing but looking, and rotate them, because looking at nothing in the rain for an hour is harder than the work.

048

“Two taps come here. Three taps run. There is nothing else.”

Rosa Kaminski, schoolteacher, day 100

WHY IT WORKS

Whatever signalling system you design, the version that gets used is the version people can perform while frightened, and that is a very short list. Two

signals is a list people can perform. Kaminski had watched twenty-two years of children under drill conditions and she knew that the vocabulary shrinks to nothing under stress, so she built for the worst state rather than the calm one. Tapping also has the right acoustics, which she did not know and got right by accident: a knuckle on wood is mid-range and directional and it does not carry past the building.

HOW IT GOES WRONG

A vocabulary of two cannot say the thing that most needs saying. There is no signal in that system for I am hurt and I am not in danger. On the eleventh of March in the second year a boy called Aaron Petch tapped three, which means run, because it was the only thing he had. The household ran. He had a broken leg under a fallen gate two hundred yards away, and they were a mile off before anybody worked it out. He was alive when they came back the next morning and he was not there. Add a third signal. Add only one, and make it mean come slowly and bring the bag.

049

“Never shout a name.”

Priya Raghunathan, A&E nurse, day 120

WHY IT WORKS

A human voice calling a name is the most effective recruiting sound there is and it beats a gunshot, which surprises people. A shot is one event, over in a fifth of a second, and half a mile away it is a thing that might have been a door. A name is a voice, pitched where voices are pitched, repeated, getting louder each time because you are getting more frightened, going on for a minute or more. It says a person is here, it says where, and it says that person is distressed and stationary. Raghunathan had watched it happen twice in a hospital car park and never shouted again.

HOW IT GOES WRONG

A rule against calling out is a rule against calling for help, and it does not distinguish. Her own group carried it too far and got quiet about everything. A woman named Ruth Colley went into a flooded cellar on Barkerend Road

after a dropped box and was in there for two days. There were four people in the house above her. They had all heard something, and had all decided on their own not to shout. Nobody in these notebooks ever mentions this. It is the failure of the good pupils. Take the rule as written for open ground, and hang the opposite of it on the inside of the door: in this building we call out, and we call out early, and we do not wait to be sure.

050

“The generator runs one hour a week.”

Jonah Kite, radio operator, day 130

WHY IT WORKS

A small petrol set is around sixty-five decibels of low-frequency noise at a couple of yards, which is not much, and it is unbroken, and unbroken is what does the damage. Everything within a mile that can hear it has all afternoon to walk towards it, and unlike a bang, it does not stop and let them lose interest. Kite worked out in the first autumn that his set was not attracting things from a distance, it was accumulating them over days, and the crowd at the bottom of the lane was the sum of a fortnight of Wednesdays. One hour a week, in daylight, in wind, with the set in a pit under a straw bale.

HOW IT GOES WRONG

Rationing the power rationed the listening, and the listening was the entire point of the machine. He was off the air for the six days that mattered most in the second year and heard the news about the Bradford road eleven days after everybody else. The bigger failure was the schedule. One hour a week is one hour a week at the same time, because that is what a schedule is. By the third summer there was a man who knew when Kite's set ran, and he had told other people. They came on a Wednesday. A ration is right. Roll the day and the hour every week, and take the small cost of being unpredictable to your friends as well.

051

“If you want a street quiet, make a noise in the next one.”

Idris Pyle, day 100

WHY IT WORKS

They follow sound and they keep following it while it lasts, and they do not check whether the sound is a person. A wind-up alarm clock in the footwell of a car with the window down will run for twenty minutes and hold a drift’s attention for an afternoon. Once the front of a crowd is moving, the back of it follows the front. No clock is needed after that. It is the cheapest tactic in this book. Pyle used it to empty the whole of a shopping parade at Hunslet, twice, and walked in behind them with a wheelbarrow, and there is no arguing with the results.

HOW IT GOES WRONG

You have not removed a crowd. You have aimed it. Whatever comes to the clock is now two streets from where it was, in a place you chose and did not look at first. On the second occasion the parade Pyle emptied was three hundred yards from the almshouses on Church Street. There were eleven people in there who had been getting away with it since the first spring. Six of them. He is the only person in this book who has written down a thing of that kind about himself, which is a point in his favour and is not very many points. Look at where the noise is going. It is going to somebody.

052

“Muffle the dog. Do not get rid of the dog.”

Duncan Ferrier, gamekeeper, day 140

WHY IT WORKS

A dog is the best perimeter anybody has and nothing built out of wire and tins comes close. It smells them at two hundred yards, upwind and in the dark, and it knows the difference between a fox and a man, which no bell has ever managed. The problem is only the barking, and barking is trainable, and

Ferrier had trained working dogs for twenty years. What he taught was a silent alert: the dog comes to you, presses its head against your leg, and holds. Two of the settlements up the valley run his dogs' grandchildren and neither has been surprised in two years.

HOW IT GOES WRONG

That training takes a calm handler, a hundred repetitions and about four months, and almost nobody in the first year had any of the three. What most people had was an ordinary frightened family dog, which barks, and a rule from a gamekeeper saying keep the dog. There are at least four households in these notebooks that were found because of a dog they had been told not to get rid of. A dog costs about the food of a small adult, every day, all year. Keep the dog if you can train it or it is already trained. Otherwise you are keeping an alarm that goes off at the wrong things and eats.

053

“Nothing is answered after dark.”

Fen Doherty, poacher, day 160

WHY IT WORKS

At night the air above the ground is warmer than the ground, and that bends sound downward and keeps it travelling, so a voice at two in the morning reaches two or three times as far as the same voice at noon. That is the physics, and the human consequence is that anybody calling in the dark has chosen the ground, knows you cannot see, and has the whole night to work with. Doherty's household answered nothing between dusk and light for three years, on any pretext, and were never once robbed, and he is a man who had a great deal worth taking.

HOW IT GOES WRONG

Real people call at night, because night is when things go wrong. A boy walked up the Elland Road side of the estate in the second February calling for anybody, for about forty minutes, in a hard frost. Eleven households heard him and every single one of them had this rule. He was found in the morning between two cars. There is no version of this where everybody follows the rule

and that boy lives. The revision people have arrived at up here has three parts. You do not answer with your voice. You do not open your door. You put a light in a window that means come to the light at dawn, and somebody watches the light all night. It is not much. It would have been enough for him.

054

“Silence is a season. It is not a rule.”

Femi Adeyemi, live sound engineer, day 300

WHY IT WORKS

This is the correction to his own first entry and to most of this chapter. A household needs to chop, build, hammer, dig, argue and sing. If all of that is forbidden then none of it happens. A group that does none of it for a year has no fuel, no repairs and no way of being a group. So you budget noise instead of banning it, and you spend the budget when it is cheapest, and in this country that means the frost. Below freezing they are standing still. Wind and rain mask. Winter is when you take the roof off and put a new one on, and there is a rhythm to the year up here now that comes directly out of this sentence.

HOW IT GOES WRONG

A budget with permission in it gets spent, and then gets spent again, and the drift is invisible from inside. Adeyemi's own settlement went from an axe in a January frost to a forge running in October in about eighteen months, one reasonable decision at a time, with nobody at any point doing anything indefensible. That is how it always goes. Somebody has to hold the number, out loud, and be unpopular. Adeyemi's note is that the person holding it should not be the person who decides what the work is. Those two jobs in one head is how the number moves.

055

“Make some noise.”

Solomon Quaye, barber, day 1,100

WHY IT WORKS

Quaye cut hair on Peckham High Street for nineteen years and his position is that a settlement of people who have whispered for four years is not a settlement, it is forty individuals standing near each other. Sound is most of how people are together. Singing, arguing across a yard, a child shouting, the noise of a room where something is being made. He started a Sunday in the frost where the rule is suspended for three hours between the wall and the barn: instruments, talking at a normal volume, and a haircut. Attendance is total. Two of the four settlements up this valley have copied it and both of them report the same thing, which is that people stopped leaving.

HOW IT GOES WRONG

It was quiet for a reason and the reason has not gone anywhere. A settlement at Guiseley took up a version of this in the second year, without the frost condition and without the walls. They were found in the autumn by people, not by them. There are nine of them left out of thirty-one. Quaye's rule has three conditions on it and they are all load-bearing: hard ground, hard frost, and a wall you have walked yourself that week. He has never once suspended it for a wedding in June, and he has been asked, and the asking is the hardest part of the job.

Water, Food, and Teeth

Nobody wants this chapter. It is the one people skip and it is the one that would have saved the most of them.

Go through eleven hundred notebooks and count what is written on the last page that anybody wrote, and a picture comes out that nobody in the first year would have believed. After the first winter, the things killing people in this county are these, in order. Cold, dirty water, teeth and jaws, wounds that were not cleaned, other people. And then, a long way down, them. That order has held for three years. There is a whole literature in these papers about how to put a spike through a skull. There are four pages in total about how to keep your mouth from killing you. The skulls are not what is in the graves.

The reason is simple enough. Every one of us came out of a world where water arrived clean, food arrived safe, waste went away and a bad tooth was an appointment. None of those were skills. They were services. They went in a fortnight. What replaced them was a set of jobs that are dull, daily, unforgiving and completely unglamorous. Nobody has ever once been thanked for doing them properly. They will not tolerate being done four days out of five. A well dug in the wrong place does not fail dramatically. It fails in August, to everybody, at once.

The last four entries in this chapter are about wounds and about bites, and I have put them here rather than in a chapter of their own for a reason. A bite is a medical event. It is not a moral one, it is not a supernatural one, and it is not the beginning of a story. It is a wound with a known course, a known timescale and a known outcome. Treating it as

anything else is how families end up doing things in barns that they cannot afterwards live with. Read those four cold. They were written cold, by people who had done it more than once, and two of them are the most careful writing in this book.

056

“Boil it. If you did not watch it boil, boil it.”

Gwen Prydderch, water engineer, day 70

WHY IT WORKS

A minute at a rolling boil finishes everything biological that this country has to offer, and biological is what kills you. All the dysentery in these notebooks. The thing that went through the Kirkstall camp in the second July and took nineteen people in nine days. Prydderch’s second sentence is the one that matters. Water that somebody says is boiled is not boiled. The failures are all social rather than technical. The person who let it off at a simmer was tired and it was late and it looked like it was going. She ran one pot, one fire, one named person a day, and that household has not had a case in three years.

HOW IT GOES WRONG

Boiling is expensive and it fixes only one of the two problems. It costs wood, which costs cutting, which is noise and exposure, and a household of six needs eighteen litres a day and that is a real fire every day forever. It does nothing to what came out of the works at Knostrop, or off the fields, or out of a tank with a dead pigeon and four years of lead in it. And boiled water in a warm store is fine for two days and is not sterile after that. It goes back to being ordinary water the moment it is standing in a bucket in a kitchen. Boil it, cover it, and use it inside two days.

057

“The latrine goes downhill of the water and further away than you want it.”

Ivy Okonjo, midwife, day 45

WHY IT WORKS

Thirty yards and downhill of anything you drink, and that number is not a superstition, it is roughly how far it takes ordinary ground to do the filtering. Everything Prydderch's entry is about gets into the water at this one point. The camps that failed in the first two years failed here more than anywhere else. They failed within about six weeks of setting up, every time, in the same way. Okonjo had delivered babies in places with no plumbing. She knew that the whole of public health, when you take the pipes away, is one hole in the right place and a rule about hands.

HOW IT GOES WRONG

A latrine thirty yards away is not used at three in the morning in February by a woman with dysentery. That is not a failure of character. It is arithmetic about a person who is ill in the dark and cold. What happens instead is behind the shed, ten feet from the door, which is worse than having no rule at all because nobody is admitting to it. Okonjo's own correction took her a year to write down. The rule has two halves and everybody publishes the first. The hole goes thirty yards downhill, and a covered bucket goes by the door. The bucket is emptied into the hole by whoever is up first, and nobody is ever asked about it.

058

“Perishables first. The tins are for February.”

Rob Hatherleigh, logistics corporal, day 20

WHY IT WORKS

It is the order every quartermaster has used and it is right for the reason that is easy to miss. You are not choosing what to eat. You are choosing what to

throw away. Anything with a short life is either eaten now or lost entirely, and a tin loses nothing by waiting a year. So the fresh goes first, then the dry, then the tinned, and the shape of the store you end up with in January is the shape you would have designed. Hatherleigh's depot came out of the first winter with four months in hand when three groups within five miles were eating candles.

HOW IT GOES WRONG

It assumes there is a February and that you are in it. Two households in these papers ate to a plan for a spring they did not reach, and left stores behind them that other people are eating now. That is not the worst outcome in the book. It is not what they were planning. The subtler failure is what it does to the autumn. Heavy work happens in September and October. The plan has you eating the thin end of the perishables then, and the good tinned fat in February when you are sitting still. That is backwards for the body doing the work. Feed the season you are in. Move about a fortnight of the good food forwards into the digging.

059

“Salt, fat, and something green.”

Ade Lawal, chef, day 60

WHY IT WORKS

These are the three things a scavenged diet lacks and each one kills differently. Without fat you can eat rabbit until your jaw aches and starve doing it, which happened to at least two people in these notebooks and which reads like nonsense until you understand it. Without salt you cannot work in July and cannot keep anything through October. Without something green you have about three months before your gums go and your old wounds open. There were people in the second March who thought they had been bitten and had scurvy. Lawal knew every one of these from the trade and he wrote the rule so that a person who could not cook could still eat correctly.

HOW IT GOES WRONG

Something green in February costs more than it returns. Going out for four hours in the cold for a handful of nettles that are not there yet is a bad trade in calories and a worse one in exposure. The answer is not the winter forage. It is the autumn one, dried and hung. His own household's failure was different and it is a professional's failure. He had them cooking, beautifully, every day, and not preserving. They came out of the second autumn with the best food anybody had eaten in a year, and eleven weeks of store where they should have had twenty.

060

“Plant more than you can eat and less than you can weed.”

Maggie Lunt, hill farmer, day 200

WHY IT WORKS

Everybody who has never grown anything believes the constraint is seed and it is not, it is labour in June. A plot that is twice what you can keep on top of does not give you twice the food. It gives you a field of weeds with some potatoes in it, and it takes the same ground and the same water. Lunt's rule sets the size by the bottleneck rather than by the ambition. The households up this valley that took it seriously are the ones with a store. The ones that dug up half a hillside in the first optimistic April fed the rabbits.

HOW IT GOES WRONG

A crop is a fixed asset and a beacon. It is visible from a distance, it cannot be moved, it announces the number of people it is meant to feed, and it ripens on a date that anybody watching can also read. Two settlements in these notebooks were robbed at exactly the right week by people who had done nothing all year but wait. Then there is the ordinary agricultural disaster. She went into the second year on potatoes, because potatoes are the calories, and the blight came through in August. She has planted four things every year since and expects to lose one of them.

061

“Learn one animal properly.”

Isaac Bramall, beekeeper, day 40

WHY IT WORKS

A person who half knows chickens, rabbits, goats and bees has four things that will die on him and no idea which. A person who knows one, in detail, through a full year, has a thing that produces food without being hunted for. It turns grass or scraps or flowers into calories while he is asleep. Bramall had kept bees for thirty years and got honey, wax for light, and a swarm to trade, out of hives he did not have to feed at all. The knowledge is the asset. Everybody can acquire the animal.

HOW IT GOES WRONG

One animal is one bad winter away from nothing, and his was the second one. He lost every colony he had between December and February, to cold and to a mite he had no treatment for. A man who has spent two years being the person with the bees turns out to have no second thing. And no time to learn one at fifty-eight. He is alive because his neighbours fed him through a spring. Know one animal properly. Then spend one afternoon a month being an amateur at a second, and be content to be bad at it. Bad at two is the position Bramall would have wanted in that February.

062

“A tooth will kill you sooner than a bite.”

Marta Kowal, hairdresser, day 240

WHY IT WORKS

Count them in the notebooks. After the cold and the water, the next thing on the list is jaws, and it is a long way ahead of anything with teeth of its own. An abscess in a lower molar has a short road into the floor of the mouth and the neck. From there it is days. There is no surgeon and no antibiotic worth the name in this county. The diet does not help: nothing sweet any more, which

people think saves them, but everything is gritty and acid and hard, and enamel goes. Kowal's regime is a rag, salt and two minutes, every night, for everybody, forever, and she checks mouths on the first Sunday of the month like a schoolteacher.

HOW IT GOES WRONG

She began pulling teeth in the second summer because there was nobody else, and she is a hairdresser, and she says so on the page in a way I have not softened. Eleven extractions. Three went badly, one of those very badly, and a woman who had a treatable abscess in March had a broken root and a worse infection in April. The rule also drives people towards the pliers too early, because a rule that says teeth kill makes every ache an emergency. Clean the mouth every day and you will not need the pliers. That is the entry. The pulling is not the technique, it is the confession of the technique having failed.

063

“Six days, in a room with a door that opens from the outside.”

Yusuf Errai, veterinary surgeon, day 75

WHY IT WORKS

Turning takes between four hours and nine days and the spread is not random. It goes with body mass and with how much got in. A small child bitten badly is hours. A heavy man with a graze is most of a week. That is the whole of the science and it is enough to act on. Errai's protocol is the one that has held up. The bitten person goes in a room alone, with water, with light, with the door bolted on the outside and a hatch. Somebody sits with them and talks to them through it. Nobody is tied up. The last part is not sentiment. Restraint has killed people in these papers who were never infected at all, of thirst, of cold, and once of a fire.

HOW IT GOES WRONG

Six is not nine. He chose six because six covers most cases and because a household cannot hold a room and a watcher for nine days. He has written,

twice, that this was a compromise with the labour available and not a finding. Two people in these notebooks came out of that room on the sixth day and turned on the seventh, and one of those took three others with him. And the protocol is unbearable in a way that no writing conveys. It is a rule for a stranger. Almost nobody has ever managed to perform it on their own child, and I am not going to pretend the failure rate there is anything other than total.

064

“Cut it off.”

Priya Raghunathan, A&E nurse, day 88

WHY IT WORKS

It is in the book because it works sometimes and because a rumour that it never works has cost lives. There are eleven amputations in these notebooks, all of them below a knee or an elbow, all within a few minutes of the bite. Three of those people are alive and none of the three turned. The reasoning is that it does not travel instantly and a limb removed above the wound at speed can take the whole of it. That is the entire basis. It is thinner than anybody would like. Raghunathan wrote it out as a procedure, with a tourniquet, a time and an order of operations. If it is going to be done at all, it will be done by somebody who is not a nurse.

HOW IT GOES WRONG

Eight of the eleven died. They died of blood loss and of shock and, two of them, of the wound going bad a fortnight later in a house with no way of stopping it. Worse than that number is what sits behind it. At least two of the eight were bitten through a coat and a shirt, and there is a real chance neither of them was ever infected. They had a leg taken off in a kitchen on a maybe. Nobody can tell in the first two minutes whether skin was broken, and the first two minutes is the entire window. Raghunathan's own note is the honest one and I will not improve on it. This is a procedure with three survivors and eight graves, and the survivors are the ones who tell you about it.

065

“Take the boring shelf.”

Nadia Rahman, pharmacist, day 300

WHY IT WORKS

By the second year every obvious thing had gone from every shop in this county and the boring shelf was untouched, and it still is. Salt, bicarbonate, vinegar, plain flour, dried pulses, sugar in ten kilo bags, TCP, sanitary goods, needles, reading glasses, boot laces, matches, and every tin of anything nobody has heard of. Rahman’s insight is about the psychology of a hurried person. People scavenge for a meal, not for a year. What survives four years of stripping is exactly the material that does not look like dinner. Half the useful stock in the Long Room came off boring shelves in the third summer.

HOW IT GOES WRONG

She spent three days on a chemist at Guiseley in the second November. It was a serious piece of work with a real risk in it. She came out with almost nothing that mattered. Most of what is in a pharmacy needs a diagnosis you cannot make, a cold chain you do not have, or a course you cannot complete. The shelf that would have paid for those three days had the salt and the plasters on it. It was in the shop next door. She walked past it twice. The boring shelf rule is right and it is much broader than she wrote it. It is not only the dull items in the interesting shop. It is the dull shop.

066

“Eat it now.”

Bea Winslow, retired postmistress, day 700

WHY IT WORKS

A store that is never drawn on is not a store, it is a museum, and Winslow has watched three households guard one into the spring while the people guarding it got thin. Her rule has two halves and both are permission. The first is that food kept past its usefulness is a loss, so the jar that is turning gets

opened on a Tuesday for no reason. The second is harder. A meal that is better than it needs to be, on a bad day, for no occasion, is not an indulgence and it is not weakness. It is maintenance on the only machine you have got. She keeps a shelf of unreasonable things and spends off it about once a month.

HOW IT GOES WRONG

Somewhere past the end of this rule is the seed corn. It has been eaten, twice that I know of, by settlements that panicked in a February that turned out to be over in eleven days. Both of them are gone. The line between drawing on the store and eating next year is not obvious from inside a hungry winter, and there is no formula for it. So Winslow does the thing that works instead. The seed and one month's food live in a different building, counted, by two people, and are not part of the decision at all. Eat what is on the shelf, freely, and put the future somewhere your hungry self cannot reach it.

III

EVERYONE ELSE

Other People

I have put off writing this chapter for two winters and I want to say why, because the reason is the argument.

Every set of rules anybody has ever pinned to a wall about strangers is a rule for getting the answer wrong in a chosen direction. That is all a rule of this kind can be. You are being asked to decide, in about four seconds, on almost no evidence, whether the person at the end of the lane is a danger. There is no method that gets that right. Anybody who tells you there is has either been lucky or has stopped counting the people they turned away. So what you are choosing is not accuracy. You are choosing which mistake you would rather make, over and over, for the rest of your life. Make that choice in daylight. The alternative is having it made for you at nine at night by whoever is most frightened.

Here is the count, for what it is worth, and it is worth something. In these eleven hundred notebooks I can identify four hundred and six encounters with strangers described in enough detail to classify. Eleven of them ended in somebody being robbed or killed by the stranger. Twenty-nine ended in the stranger being killed or driven off by people who turned out to have been in no danger at all. The rest were nothing: a wave, a trade, a night in a barn, a piece of information, a person who walked on. If you are going to be wrong, and you are, the numbers say which way. You are far more likely to harm somebody who came to you for help than to be harmed by them.

That is not an argument for opening the door. Eleven is not nothing, and two of the eleven took whole households. It is an argument for

knowing what you are trading, and for not letting the fear do the arithmetic. The fear has been wrong about the size of the risk for four years running, and in this county the wrongness has been fatal.

The entries here disagree more sharply than anywhere else in the book. Rule 70 says the door does not open, and the man who wrote it has buried people who opened theirs. Rule 77 was written by a woman who has opened hers every time she has been asked, for four years, and who is alive, and who is not naive, and whose settlement is the largest in this valley. Both of them are here in full and I am not going to arbitrate between them.

067

“They have decided about you before either of you has spoken.”

Idris Pyle, day 200

WHY IT WORKS

Pyle spent his working life on the first four seconds and he is right about what is in them. Before a word, the person opposite has read five things. Your condition, which is how well you have been eating and sleeping. Your posture, which is whether you expect trouble. What is in your hands. Whether you are looking at theirs. Whether you are alone. All of that is legible at forty yards and none of it can be talked away afterwards. His practical point is that you may as well choose what it says. Stand square, hands visible and empty, look at the face and not the hands, and stop walking before you are asked to.

HOW IT GOES WRONG

He is describing a performance, and a performance costs something to run and can be read as a performance by anybody who has met one. There is a particular flatness that very careful people acquire, and in the third year it has become its own signal in this county, and it is not a reassuring one. Pyle also had four years of this and ended up unable to meet anybody without working, which is a way of being profoundly alone in a crowd. Do the hands and the stopping, which are courtesies and cost nothing. Leave the rest of it. Most people are not reading you nearly as closely as he was.

068

“Meet them at a distance you can both walk away from.”

Marcus Okon, police sergeant, day 210

WHY IT WORKS

Forty yards of open ground with a clear line out behind each of you is a geometry that lets a meeting end in nothing, and most meetings should end in nothing. Close the distance and you have taken that option off the table for both of you: now something has to be resolved, and it has to be resolved by two frightened parties standing within reach of each other. Okon ran gate meetings on this principle for two years without an incident. Space is not coldness. Space is what makes it possible for the other person to change their mind and leave without losing face, and losing face is what most of the eleven bad outcomes in these notebooks are actually made of.

HOW IT GOES WRONG

Forty yards of open ground reads as hostility to a family that has walked eleven miles with a child, and it reads that way because it is, a little. There is a woman in Farsley who will tell you about being held at that distance in the rain for twenty minutes while a good, careful man decided about her. It ended her willingness to ask anybody for anything, and it has not come back. And the distance that is safe from a person is the distance a rifle prefers. Okon's own amendment is that the geometry is for the first minute and not the twentieth, and that somebody has to be willing to close it, and that it should be you.

069

“Trade first. Talk after.”

Tomas Vrba, lorry driver, day 150

WHY IT WORKS

An exchange of two small things establishes more in ninety seconds than an hour of conversation, and it does it in a way nobody can fake, because a trade

is a claim that gets tested immediately. It says: I have a surplus, I would rather have a relationship than your possessions, and I expect to see you again. Vrba traded salt in half-ounces up and down this valley for two years, and built out of it a network of eleven households that fed information to each other. He did it without once having a conversation about trust. It is a word that does no work at all.

HOW IT GOES WRONG

A trade shows your hand. Salt in half-ounces says you have a lot of salt, and by the second summer everybody between here and Otley knew which man to rob if you wanted salt, and two of them tried. It also sets a price on your household, and a household with a known surplus is a fixed asset in the sense of chapter six. Vrba's fix is the one the old markets used: trade at a third place, on a day, in small quantities, and never at either party's home. He has not been robbed since he started doing that, and he says the network works better because everybody has to walk.

070

“The door does not open. There is no version where the door opens.”

Rob Hatherleigh, logistics corporal, day 240

WHY IT WORKS

He has the eleven cases and he is the only person who has written them all up. The pattern in them is consistent and it is worth having in front of you. In nine of the eleven, the door was opened on purpose, to somebody who was asking for help, by somebody who had decided this one was different. In seven of those nine the person asking was in real distress, was alone, and had walked a long way. They were also being followed by four people who had sent them. Hatherleigh's argument is that a rule with an exception in it is not a rule, because the exception is the only door anybody has ever come through. Food goes out. People do not come in. He has run it for three years and lost nobody.

HOW IT GOES WRONG

Twenty-nine against eleven. That is the number from the top of this chapter and it is the whole case against him. He has read it and does not dispute the arithmetic. He says the twenty-nine are not his responsibility and the eleven are. What his household has become in three years is the part he does not write about and other people do. They are fed. They are safe. They are eight people who have not admitted anybody since the second spring. Three have left. The ones who left all left for the same reason and said so. There is a cost to being the household that everybody in the valley knows will not open. It does not show up as a death. It shows up as nobody coming when it is your turn.

071

“The dangerous ones are not hungry.”

Callum Reith, day 1, 100

WHY IT WORKS

Read the condition and not the manner. A desperate person is frightened, unpredictable and mostly harmless. Desperation is loud and slow, and it wants food rather than a fight. Almost every person who has frightened a household in these notebooks by shouting at a gate turned out to be exactly what they looked like. The ones who took whole settlements were fed, rested, unhurried and polite, and they were fed and rested because they had been doing it for a while. Reith’s rule inverts the instinct, which is to fear the wild-looking man, and the instinct has been wrong nearly every time.

HOW IT GOES WRONG

It is a heuristic and he used it as a licence. A group that is well fed and organised is, more often than not, a working settlement that has planted something. Reith spent his seventeenth year in a crew that used exactly this reasoning to choose who to take from. He has written about it. I have printed the entry with the rule that came out of it. That is the failure of this rule: it identifies the strong, and there are two things you can do with that identification, and one of them is what he did. He is twenty now and he grows things. He asked me to leave this in.

072

“Children are not a trick.”

Ivy Okonjo, midwife, day 190

WHY IT WORKS

The rumour that children were being used as bait went through three counties in the first autumn, and it did more harm than almost anything else on the air. Okonjo went looking for the cases. She found two, in eleven hundred notebooks, in four years. Set that against what the rumour cost. Households that would not answer a child at a gate. A boy at Elland Road who is in chapter five. And a lasting willingness to treat the most vulnerable person in front of you as the most likely threat. The base rate is the argument. It is not sentiment, it is counting, and she did the counting.

HOW IT GOES WRONG

It happened twice. Both cases are documented, both were in the first eighteen months, and in one of them a household of six was finished by it. Okonjo does not hide either case and her position is that a rule which is right ninety-nine times in a hundred is still a rule that gets somebody killed, and that anybody adopting it should know they are accepting that. What she does in practice is the thing to copy. The child comes in. The child is fed. One person stands where they can see the road for the next two hours. Nobody makes the child answer questions about who is with them.

073

“Lie about the number. Never lie about the food.”

Idris Pyle, day 260

WHY IT WORKS

His argument is that a household's apparent size is the main thing deterring anybody, and that it cannot be verified from outside in a short observation. The food can be. So you inflate the one, tell the truth about the other, and keep your credibility where it matters. He ran it for a year. It sounds right and

it is beautifully put. It is one of the two rules of his that I have marked in the note at the back. The evidence runs the other way, and I want it in the book with the correction attached rather than left out.

HOW IT GOES WRONG

Anybody serious watches for two days, and two days of watching gives them your real number, your rota and your water run, and now they know you lie. Worse is the thing Pyle did not model. A large household implies a large store, and in the second winter the two households in these papers that had most successfully inflated their numbers were both attacked, and both were attacked because they looked big enough to be worth it. The rule optimises for deterring one man walking past and it advertises to the four men who are planning. Tell the truth about the number and let it be small. Small and known to be defended is a much better position than large and interesting.

074

“Ask them for something small.”

Ruth Vellacott, day 220

WHY IT WORKS

Vellacott's people have been doing versions of this for three hundred years, and it works on something old in us. A person who has done you a small favour is better disposed towards you afterwards. The effect is larger than the favour. So you do not open with an offer, which puts them in debt and makes them wary, and you do not open with a large request, which is a demand. You ask for water, or the road, or the time of the tide. It is nearly free for them, it gives them the position of the person with something to give, and it turns a confrontation into a transaction that has already gone well once.

HOW IT GOES WRONG

It marks you as somebody who asks, and in a bad winter that is a category people are watching for. There is also a real cost hidden by the word small. A household that has nothing is being asked to demonstrate that it has nothing, in front of a stranger. Vellacott has been on the receiving end of that and describes it as one of the worse things that has happened to her. Ask for

something they obviously have. Directions, water in a place with water, a light. Never ask a thin household for food and call it small.

075

“Never show them the way home.”

Fen Doherty, poacher, day 175

WHY IT WORKS

Everything about your household that matters is its location, and location is the one thing that cannot be undone once it is given. Doherty meets people on a lane half a mile from where he lives, walks doglegs, comes back after dark by a different way, and has never in four years let anybody follow him in. It costs him an hour a week. In exchange, the eleven encounters in this chapter that ended badly all had one thing in common. Somebody knew where the house was. No defence, no wall and no rota competes with not being findable.

HOW IT GOES WRONG

He is alive and his household is four people and it was nine. Nobody has ever come for them and nobody has ever come to them either. When the roof went in the second February there was no one to ask. When his sister was ill there was no one to ask. The settlements that grew in this valley grew because people knew where they were. Doherty’s method is the correct method for a household that intends to stay four people forever. It is not a method for building anything, and he is the first to say so, and he has said it to me twice in a way that I have thought about since.

076

“Give the first stranger of the winter something.”

Maggie Lunt, hill farmer, day 300

WHY IT WORKS

Reputation travels faster than anybody expects and it is the cheapest defence there is. What Lunt wants said about her place, up and down this valley, is two sentences: they will give you a meal at the gate, and it is not worth trying anything. Both halves are necessary and the first is the one people neglect. A place known to give a little is a place nobody needs to raid, because raiding is dangerous and asking is not. In four years she has fed perhaps sixty people at that gate. She has been approached with any other intention exactly once.

HOW IT GOES WRONG

It produces a queue, and a queue is a fixed asset with a number on it. In the second winter she had forty people at that gate over eleven weeks, and she could not feed forty people. The ones she turned away went off with a much worse story than if she had never fed anybody at all. Generosity that runs out is remembered as a refusal. Her fix is to set the amount for the season in advance, publish it, and hold it. A meal at the gate and a night in the byre. That is what it is and what it has always been. Nobody who arrives in February is told something different from the man who arrived in October.

077

“Open the door.”

Sister Agnes Moray, day 400

WHY IT WORKS

She has opened it every time she has been asked, for four years, and her settlement is the largest in this valley and it is not an accident. The reasoning is not what people assume. It is not that everybody is good. It is that a settlement is made of people who arrived. Every single person now feeding her, mending her roof and standing her watches was at some point a stranger

at the gate. A place which does not open cannot grow. A place that cannot grow is on a slow clock, and it runs out somewhere around the fifth or sixth winter, when the people in it get old. The twenty-nine against eleven is her arithmetic. She got there before I did.

HOW IT GOES WRONG

It has cost her. Twice. The second time, in the third autumn, it cost her four people and most of a store. She opened the door again eleven days later. There are people in that settlement who have never forgiven her and are still there. What makes it work is not the opening, which is the part that gets quoted. It is the four things underneath it, and she lists them every time and nobody ever writes them down. The door opens into a yard and not into the house. Somebody is always standing where they can see the yard. Nobody sleeps inside the walls on their first night. And the decision to open is hers alone, so that it can never become a thing the household argues about at the gate with a person listening. Open the door. Build the yard first.

The Group

Nearly every settlement that has failed in this valley failed from the inside, in the autumn, over something small.

I did not expect that when I started sorting the notebooks and I have had to change my mind about it twice. The picture from outside is always the same: a place was overrun, a place was raided, a place had bad luck. Read the papers from inside and the overrunning turns out to be the last event in a sequence that started nine months earlier, with two people who stopped speaking, and a rota that everybody knew was unfair and nobody would say so. A decision that got made three times because it never got made properly once. By the time anything with teeth arrived there were four people doing the work of eleven and nobody on the north wall. The thing that finished them would not have got within half a mile of the place it had been in March.

So this chapter is about governance, which is a heavy word for what it is. It is a set of answers to about six questions. Every group of more than four people has to answer them, whether it notices or not. Who decides. How the unpleasant work is shared. What happens to somebody who cannot work. What happens when somebody wants to leave. Who counts the food. What is done when a person dies. Groups that answered those out loud, early, in writing, while nothing was wrong, are with one exception still here. Groups that left them to be settled by whoever was loudest in the moment are, without exception in these papers, not.

One thing before the entries. Every rule here was written by somebody who had watched a group come apart. None of them was written by

somebody who had built a good one from nothing, because the people who did that were busy. There is a bias in this material towards the diagnosis and away from the cure, and I have no way of correcting for it. What I can tell you is that Moray's settlement is the largest and healthiest thing anybody in this county has made. It does eight of these eleven things. It has never written any of them down and could not tell you which eight.

078

“Decide how you will decide, before there is anything to decide.”

Rosa Kaminski, schoolteacher, day 150

WHY IT WORKS

A group in a crisis can do one thing at a time. It will either resolve the crisis or resolve the question of who is allowed to resolve it. It cannot do both. Kaminski's group spent one wet afternoon in the second June on a single sheet of paper. These four things are decided by whoever is on watch. These three by the whole household in the evening. This one by two named people together. And here is what happens if those two disagree. It took ninety minutes. It has been used perhaps four hundred times since and it has never once been argued about, because the argument happened in June when nothing was at stake.

HOW IT GOES WRONG

Rules written in the calm are written by the people who are comfortable in the calm, and they are almost always the articulate ones. What a wet afternoon with a sheet of paper produces is a system that suits people who like wet afternoons with sheets of paper. Kaminski's own document gave three decisions to whoever was on watch and did not notice that the watch rota put two particular men on watch for most of the winter nights. The paper needs revising once a year, by somebody who was not in the room the first time. It needs to be revised while nothing is wrong, which is the exact condition under which nobody wants to.

079

“Every rule has a name on it.”

Nell Okafor, librarian, day 200

WHY IT WORKS

An anonymous rule cannot be argued with, because there is nobody to argue with, and so it becomes furniture and then becomes doctrine. Okafor made everything in the Long Room carry the name of whoever proposed it and the date, and the effect was immediate and not what she expected: bad rules got repealed. It turns out that people will not challenge the way things are done, which is impersonal and enormous. They will happily say that they think Nell’s rule about the lamps is wrong. That is small and specific and it has an obvious remedy.

HOW IT GOES WRONG

A name on a rule turns repealing it into a defeat for a person, and people defend rules to defend themselves long after they have stopped believing in them. Okafor’s rule about the lamps was wrong within a fortnight and stayed for eleven months. She is the one who wrote it and she is the one who kept it. She knew. The correction is a small piece of theatre and it works. The person who proposed a rule is the person who is asked, once a season, whether it should stay. Answering no in that particular chair is the most respected thing anybody in this building can do.

080

“The rota is the government.”

Rob Hatherleigh, logistics corporal, day 180

WHY IT WORKS

Whoever chairs the evening meeting has the visible power and it is not where the power is. The power is in who empties the latrine bucket, who takes the four-in-the-morning watch in January, who goes on the water run when it is raining. And whether those are the same three people every week.

Hatherleigh's argument is that a group's real constitution is its rota, and that everybody in the group already knows this. A settlement where the unpleasant work is visibly shared can survive almost any amount of disagreement about anything else. He built rotas for a living. His are fair to a degree that nobody has ever faulted.

HOW IT GOES WRONG

They are also joyless, and a rota that is perfectly fair and never varies turns eleven people into eleven functions, and does it slowly enough that nobody can name the week it happened. Hatherleigh's household in the second year was the best run and least happy thing in this book. The three people who left it left for a place that was worse organised and warmer. Fairness is necessary and it is not sufficient. The fix costs nothing. Let people swap. Let people volunteer for the bad shift and be thanked out loud. And put one thing on the rota that is not work.

081

“Do not make the strongest person the leader.”

Priya Raghunathan, A&E nurse, day 210

WHY IT WORKS

The first year selected hard for one skill and it was the wrong one. Being good at the dangerous part is a specific and valuable competence. It has almost nothing to do with allocating food across a winter, judging a stranger, or holding a group of frightened people together in February. Putting the same person in charge of all four is how you get a settlement that is superb in a crisis and cannot organise a harvest. There is a second reason and it is practical. The strong one is needed at the wall, and a leader who is at the wall is not available for the eleven decisions that get made while they are out.

HOW IT GOES WRONG

There are weeks when the group needs the strong one to lead and everybody knows it. A rule that forbids it produces a fiction, where the nominal leader relays what the strong one has said. Raghunathan has watched that and calls it the worst of both. The other cost lands on the person doing the dangerous

work. They take the risk and they are excluded from the decisions. By the second winter that is a specific and legitimate grievance, and in one settlement in these papers it turned into something much worse. Separate the roles, name both out loud, and give the dangerous one an absolute veto over anything that sends people outside.

082

“Let people go.”

Solomon Quaye, barber, day 500

WHY IT WORKS

A group that cannot be left is not a group, it is a situation, and everybody inside it can feel the difference even when nobody says so. Quaye's position is that the right to walk out is what makes everything else voluntary. A person who stays when they could go is a different and better colleague than a person who stays because the alternative has been made impossible. His settlement gives anybody leaving three days of food, the loan of a coat and no argument. Four people have used it in three years. Two of them came back, and both say the going was what let them come back.

HOW IT GOES WRONG

An open door bleeds in the good months. People leave in May, when everything is easy and somewhere else looks better. They are missed in November, when the roof needs doing and there are four fewer pairs of hands than the place was built around. And what leaves with them is the map: where the store is, how many are on the wall, which night the watch is thin. Quaye's household accepts all of that and has decided it is the price. The one thing he does insist on is that nobody leaves in the fortnight after a death, or in a hard frost. Both of those have been argued about. He has held both.

083

“Two people count the store, and they are not friends.”

Bea Winslow, retired postmistress, day 260

WHY IT WORKS

Winslow spent forty years in a job where the money was counted twice by two people every single day, and she understood exactly what that practice is for, which is not catching thieves. It is making theft unnecessary to worry about. A store counted in the open, by two people who do not socialise, on a known day, removes an entire category of suspicion from a group's life. Suspicion about food is the thing that ends settlements. She has never found a discrepancy she could attribute to a person. That is the system working, not the system being unnecessary.

HOW IT GOES WRONG

It says out loud that the group expects to be stolen from. In a small household that is a hard thing to have said. Two settlements in these notebooks abandoned the practice within a month for exactly that reason, and both regretted it. The other failure is Winslow's own and she tells it against herself. The count found nothing for two years and then found a shortfall of eleven pounds of oats. There was a fortnight of quiet horror. It turned out to be her own arithmetic in a cold room in December. Count twice. Then have a rule, written down in advance, about what happens on the first discrepancy, because the first discrepancy will be a mistake and everybody will behave as though it is not.

084

“Bury them properly, and everybody stops work.”

Wes Ordish, undertaker, day 130

WHY IT WORKS

Ordish buried people for thirty-one years and his claim is one that the notebooks support without much doubt. The settlements that stopped

holding funerals came apart within about six months, every time. The ones that kept them did not. A funeral does nothing for the dead. It is a machine for telling the living three things. That a person who was here mattered. That a death will be marked when it is theirs. And that the group is a thing that owes its members something. Stopping work is the load-bearing part. A burial that people attend in shifts, around the rota, says the opposite of what it is meant to say.

HOW IT GOES WRONG

A burial party is the most exposed a settlement ever is. Everybody in one place, outside the walls, occupied, upset, not watching, at a time and place that has been known about for a day. A settlement at Rawdon lost three people at a graveside in the second August, which is one of the worst things in these papers. Ordish's own practice by the third year was to bury at first light in the frost, with two people at distance who do not attend, and to hold the service afterwards and inside. He has said that separating the two was the hardest professional decision of his life and that he would do it again.

085

“Nobody alone. Nobody in more than three.”

Ewan Tait, mountain rescue, day 130

WHY IT WORKS

One is no rescue. A person alone who breaks an ankle two miles out is a death, and Tait had spent nineteen years walking towards the consequences of that. Four or more is loud, slow, hard to hide, and produces the thing where everybody assumes somebody else is watching. Three is the number that works: one doing the task, one watching the ground, one spare pair of hands who can go for help or carry. It is also small enough to lie down in a ditch. Every mountain rescue team in this country arrived at some version of three by about 1960 and none of us should have had to rediscover it.

HOW IT GOES WRONG

In a household of eight, three out is more than a third of your strength gone for a day, and if that day goes wrong you have lost nearly half of everything.

There is a subtler problem that finished a settlement at Yeadon. Three people out for eleven hours make decisions together, on the ground, with information the household does not have, and they come back having committed to something. Do it four or five times and you have a group inside the group, with a shared history the others do not share, and the household is now two things. Rotate who goes. Never send the same three twice running.

086

“Teach somebody else your job.”

Yusuf Errai, veterinary surgeon, day 320

WHY IT WORKS

Every settlement is a list of single points of failure and nobody writes the list down. Bramall’s bees. Cattermole’s canal. Kowal’s teeth, and Errai’s own, which is the whole of the medicine for four settlements and is one man of sixty-one. The rule is not about generosity. It is about arithmetic. A household whose only water engineer is one person has a fifty percent chance, over a few years, of losing water engineering altogether. The cost of preventing that is an hour a week for a year. Errai has taught his second three times over and has stopped being irreplaceable, which he describes as the best fortnight’s work he ever did.

HOW IT GOES WRONG

Teaching costs the expert their time and, if they let it, their standing. There is a real and unattractive way in which being the only person who can do a thing has been keeping some of these people fed. Worse is the half-taught apprentice, which is Kowal’s chapter six entry in another form. Somebody with four months of instruction and complete confidence is more dangerous than somebody with none. They will attempt the thing rather than wait. Teach the second. Then write down, and make them say out loud, the three procedures they are not to attempt without you, and expect that list to be the thing they resent.

087

“Somebody has to be allowed to be useless for a week.”

Sister Agnes Moray, day 330

WHY IT WORKS

People break. Not dramatically and not for good. The pattern is the same one Moray saw for twenty years before any of this. A person who is given a week to be no use at all comes back and works for another three years. A person who is not given it stops working properly some time in the following autumn, and never starts again. Her settlement holds about a fortnight of slack in the year for this and treats it as a stock, like food, that is laid in and drawn down. It is the single most unusual thing about that place and I think it is why it is the largest.

HOW IT GOES WRONG

Slack is bought with somebody's labour, and the somebody is always the same three people. If the week is not rationed it gets drawn by whoever is loudest about needing it, rather than whoever is closest to breaking. Moray's own difficulty is that she has never once refused it and has been taken advantage of twice by the same man. And a household of five cannot do this at all. The arithmetic does not close. Everybody knows it does not close. Pretending otherwise produces a rule that exists on paper and is never used, which is worse than not having it. Below about eight people, this entry is not available to you, and that is a real argument for being more than eight.

088

“Go, if it is that kind of group.”

Callum Reith, day 1, 180

WHY IT WORKS

He was in a crew for most of two years that took food off settlements, using the reasoning in his chapter seven entry. He left at nineteen with nothing, in September, and walked forty miles. His rule is the counterweight to

everything else in this chapter, and it needs saying because loyalty has been doing a great deal of unexamined work in these papers. A group is a tool for keeping people alive. When it stops being that and becomes a thing that keeps other people from being alive, staying in it is not loyalty and leaving it is not betrayal. Every argument you will hear to the contrary will be made by somebody who benefits from you staying.

HOW IT GOES WRONG

Almost every solitary death in these eleven hundred notebooks begins with somebody walking out of a group. That is the flat fact and Reith knows it better than I do. The ones who left in a temper, in the wrong season, without a destination, are a category. It is a large category. He got across those forty miles in September in good weather, and has said plainly that in January he would not have. So: go, if it is that kind of group. Go in spring or early autumn. Go towards a named place. Take three days of food and tell one person you trust which way. And if none of those four conditions can be met this month, plan it for the month when they can. In the meantime, do not let anybody persuade you that the deciding is settled.

The Long Middle

Nobody wrote a rule for this part, because nobody expected to be in it.

Everything in the first eight chapters was written by people in an emergency, for people in an emergency. An emergency has a shape everybody understands without being told. It is short. It is survived rather than lived in, and the whole of your behaviour inside it is permitted by the fact that it is going to end. What none of us had any preparation for is the discovery that it was not going to end. The notebooks fall silent at almost exactly the eighteen month mark. This was not a period of difficulty between the world and the world coming back. It was simply the world now. And the habits that had kept us alive through a bad winter were, carried into a fourth year, going to hollow us out and leave a county full of thin capable people who could do everything except want anything.

I have read four hundred notebooks that stop for that reason. They do not stop at a death. They get shorter, the entries get further apart, the handwriting stays perfectly steady, and then there is a run of blank pages and then nothing, and the person is alive, and I have met eleven of them. That is what the long middle does. It is the only chapter in this book where the failure mode is not a body.

So these eleven rules are different from the rest and I want to be honest that some of them will look like luxuries. Planting a tree, teaching a child to read, holding a wedding, walking three weeks to look at the sea. None of that keeps anybody alive this Tuesday. Every one of them costs labour that could go into the wall or the store. The argument for them is that

they are the only things anybody has found that keep a person interested in being alive next Tuesday. Interest in that is not a decoration on survival. It is the load-bearing part. The four hundred silent notebooks are the evidence.

This is the chapter people ask me for. When somebody comes to the Long Room and wants the book, they have almost always heard about one of these eleven and not about any of the other eighty-eight. I have thought about why. I think it is this. The rest of the manual tells you how not to die. This part is the only part that says anything at all about what the not dying was supposed to be for.

089

“Count the ones still walking. Then count them again next year.”

Yusuf Errai, veterinary surgeon, day 1,250

WHY IT WORKS

This is the first piece of good news in the book and it has taken four years to be able to write it. They rot. Slowly, and much more slowly than anything else does, and not at all in the way a body rots. But they do. The fourth year is when it began to show. Hips going, tendons at the ankle failing, the ones that come down the lane now doing so with a gait that would not have got them across a field in the first summer. Errai has counted the same three miles of road every March for four years and the number has fallen each time, and the fall between the third year and the fourth was the largest. This is measurable, it is repeatable, and anybody can do it.

HOW IT GOES WRONG

The number is falling and the danger is not. What is being removed from the population is the weak. What is left is a selected remainder: the ones that turned in the cold, that have spent four years in shade and damp, that are intact. Errai is careful about this and everybody who repeats him is not. Then the second thing, which he thinks is the more important and which nobody wants to hear. Any unburied cluster of a few hundred is still a few hundred. A hard winter preserves them. The arithmetic of a second wave has never once

required anything new to happen. Count them. Publish the count. Do not build anything on the trend that you could not defend if the trend stopped.

090

“Teach them to read.”

Rosa Kaminski, schoolteacher, day 1,090

WHY IT WORKS

It goes in one generation. That is the fact that ought to frighten people more than it does. A skill that took this country four hundred years to establish is gone in about fifteen if nobody teaches it. The children who were four when it started are eight now. There is a window and it closes. Everything in this book is a written thing. The whole method of the Long Room, and of Skipton, and of every settlement that has learned anything from another settlement’s dead, runs on paper, and paper without readers is insulation. Kaminski runs two hours a day, six days, for nine children, and she has been doing it since the second winter and she has never missed a day for weather.

HOW IT GOES WRONG

Two hours a day for nine children is two hours a day for one adult and nine sets of hands. In a settlement of thirty that is a real subtraction from the harvest. It is paid by everybody and it benefits the parents. There are settlements that have had this argument badly. The other cost is quieter and Kaminski raises it herself. Whoever teaches chooses what is taught, and there is nobody to check her. She has been the only source of what nine children think happened to the world. She has started having Quay take one hour a week for exactly this reason, and she says he tells it differently, and that this was the point.

091

“Plant a tree.”

Maggie Lunt, hill farmer, day 800

WHY IT WORKS

An apple takes seven years to be worth anything and that is the entire argument. Everything else a settlement does is inside one season. You plant in April to eat in September. You cut wood in January to burn in February. A life made only of those has a twelve month horizon, which is a survivable way to live and not a way to build anything. A tree is a wager on being here in seven years, made in public, with your hands. Lunt put in eleven in the second spring and says the change in the household was immediate and had nothing to do with fruit.

HOW IT GOES WRONG

It is a commitment to a place and it is the strongest one available. This is the chapter four problem in its most seductive form. The household with the orchard is the household that will argue for staying when it should be arguing about the water. Her own first eleven went into the flat ground by the beck because it was the easiest digging. It is ground that floods. She lost nine of them in the second winter and did it again on the slope, where the work is four times as hard. Plant the tree. Plant it where you would build, not where it is easy, and plant it in the third year rather than the first, when you know what the ground does in February.

092

“Marry people.”

Sister Agnes Moray, day 1,000

WHY IT WORKS

Her position is that a public promise, witnessed, is a piece of machinery and not a sentiment. A settlement is built out of them. They turn private arrangements into things the group has a stake in. They give everybody a role

at somebody else's important day. And they are the only mechanism anybody has found for making a commitment that is harder to walk away from than the walking away is easy. She has done eleven. She will tell you, and the notebooks bear her out, that the settlements with weddings in them are the settlements with children in them, and that this is not the obvious causation but the other way round.

HOW IT GOES WRONG

It builds households inside the settlement. Households have interests, and interests become factions. By the third year there was an argument at Moray's place about whose child got the schooling hours, and it had nothing to do with schooling. There is also the question of what happens when one of the eleven ends, which two of them have, and there is no machinery for that at all and she is making it up. She has refused two weddings in four years, both times because she judged one of the pair was asking for shelter rather than for a marriage. Both times she was told she had no right. She thinks now that she was correct once.

093

“Learn a trade that is not survival.”

Marta Kowal, day 900

WHY IT WORKS

For three years everybody in this county did the same nine jobs and the nine jobs were all versions of not dying. What has changed in the fourth year is that there are now a handful of people who make one thing well. A man at Horsforth who does boots. A woman at Bramhope who is a potter. Kowal herself, who started as a hairdresser, pulled teeth, and has spent two years learning to do it properly from a book and from Errai. A settlement that contains a cobbler is a different kind of place from one that contains eleven interchangeable survivors, and the difference shows up in what people will walk twelve miles for, which is the beginning of everything else.

HOW IT GOES WRONG

A specialist is a person who cannot be sent to the harvest, and in a settlement of twenty that is felt. There was a bad fortnight at Bramhope in the third September over precisely this and the potter went back on the rota and has done less good work since. The deeper fragility is that a trade takes years and the person carrying it is one person, which is Errai's chapter eight entry again, and Kowal has one apprentice and needs three. Her own note is worth quoting. She says the hardest part was not learning the teeth. It was being allowed to spend the time, and she was allowed it because the alternative was watching people die of it.

094

“Go and look at the sea.”

Hana Sørensen, marine engineer, day 1, 200

WHY IT WORKS

Sørensen's argument is that four years of errands is not a life, and that a journey with no cargo in it does something no amount of rest does. She is talking about a specific thing she watched happen to people. The horizon shrinks to the length of the water run and then does not come back. There is a practical half as well. Nobody in this valley knew anything about the coast until people started walking to it. What came back was worth the walking. There are working boats out of the Humber. The fishing is extraordinary. A whole different apocalypse happened out there and it has produced a whole different set of rules. Three weeks. Two people. Nothing carried but food.

HOW IT GOES WRONG

It is three weeks of exposure for nothing material, in a world where three weeks of exposure is how most of the names in this index ended, and the honest version of the rule has to say that. And Sørensen went, in the fourth May, with a man called Bell who came back alone in June with a letter. She is not dead. She is on a boat out of Paull with eleven other people and she is not coming back. The letter says why. It is the best thing in the Long Room and I have read it more often than I should. Go and look at the sea. Understand before you go that some people who look at it do not return. That this is not

always a tragedy. And that whoever is waiting for you will not be able to tell the difference for a long time.

095

“Keep a bad map of the whole county.”

Tomas Vrba, lorry driver, day 1, 150

WHY IT WORKS

What matters in the fourth year is not terrain, it is who is where. Vrba keeps a sheet with about forty marks on it: this settlement, that many people, they have salt, they need seed, this one has a smith, this one had a bad winter and will not answer. It is wrong in a dozen places at any moment and it does not need to be right, it needs to exist, because it turns forty isolated households into something that can move a surplus to a shortage. Three settlements got through the third February on grain that Vrba knew was sitting in a barn eleven miles from people who had none.

HOW IT GOES WRONG

It is precisely the document you would want if you were planning to take things from people, and Vrba carries it on his person and has thought hard about what happens if he is stopped. He has been, once, and talked his way out of it. The second failure is subtler and it is the one Doherty warned about from the other direction. A map like this makes its keeper important. There is now one man who knows the shape of this county and nobody else does. If he goes into a ditch in February, forty households go back to knowing nothing. He is teaching it to two people. He has been teaching it to two people for a year and he is not very good at letting go of it.

096

“Let the children be bored.”

Ivy Okonjo, midwife, day 1,300

WHY IT WORKS

The generation that is growing up in this is frighteningly competent. Reith is the obvious case and he is not unusual. They are quiet, they read ground, they are calm about things that reduce adults to uselessness. Every one of them has been useful since they were six, because there was no version of the last four years in which a child could be spared. Okonjo's finding, and she has had more of these children through her hands than anybody, is that the competence has come out of something, and what it has come out of is play. She has written a rule that says a child gets a portion of the day in which nothing is required of them and nothing is taught to them, and she defends it against everybody, including the children.

HOW IT GOES WRONG

It is bought with adult labour and everybody can see the price, and in a settlement of fifteen with a wall to finish there is a version of this argument that Okonjo loses and should lose. And there is a harder objection which comes from the children themselves. Reith's generation is competent because nobody was ever bored. Several of them have said, in various ways, that they would not trade it. Being needed at eight was the thing that made the first winter survivable for them, rather than the thing that damaged them. Okonjo's answer is that she is not arguing about the first winter. She is arguing about the eighth, and about what happens to people who have never once been permitted to be no use.

097

“Say their names on the day.”

Wes Ordish, undertaker, day 1,000

WHY IT WORKS

Grief that has nowhere scheduled to go turns up on its own account, in February, in the middle of a task, and it takes people out for a fortnight at a time and they cannot say why. Ordish’s whole trade was built on the observation that the marking has to happen and that the only question is whether it happens on a day you chose. So the fourth of April, every year, everything stops. Every name this settlement has lost is read out, in order, by somebody different each time. There are a hundred and six of them. It takes most of an hour. Nobody has ever suggested shortening it.

HOW IT GOES WRONG

It is a bad day. It is the worst day of the year in every settlement that does it. It costs a working day at exactly the point in April when there is work. Two people at Moray’s place leave for the week every year and have done since the second one. Ordish does not pretend otherwise and does not think the cost is avoidable. He has made one refinement in three years. The reading happens in the morning and the afternoon is given over to a meal. He found that a day which ends in the list ends badly, and a day that begins with it does not.

098

“Stop calling it the apocalypse.”

Solomon Quayle, barber, day 1,320

WHY IT WORKS

Language of emergency licenses behaviour of emergency, for as long as you let it. Quayle’s point is that a settlement which still describes its situation as a catastrophe in progress will keep making catastrophe decisions. No long work. No repairs beyond the winter. No trees. Nothing that assumes a year after next. He started calling it the fourth year rather than the fourth year of

anything, and made a small stubborn point of it in every conversation. He says it took about eight months to spread. The visible effect was people mending things properly instead of patching them.

HOW IT GOES WRONG

It normalises what should not be normal. There is a real objection here that Moray makes and that I have not been able to dismiss. A hundred and six people are dead on that list. The thing that killed them is still walking about outside the wall. A language in which all of that is the weather is a language that has given up on something. There is also the chapter two problem. Some people are running on the belief that this is temporary, and it is load-bearing for them, and Quay's sentence takes it out from under them in public. He now says it only to people who have been here four years. He has never once said it to anybody who arrived in the last twelve months, and that is the rule underneath the rule.

099

“Put the book down.”

the Long Room, the fourth winter

WHY IT WORKS

I have spent two years on this and it is the last entry and it ought to be honest about what the other ninety-eight are worth. Nobody who survived the first year had this book. They had judgement, and luck, and mostly luck. What I have assembled here is ninety-eight rules written by people who were making it up, and the index will tell you how many of them are still here. I have no way of knowing whether the rules are why the others are alive, or whether the alive ones had more time to write. A manual is a substitute for judgement and it is a poor one. The purpose of reading this is to stop needing to. If you have got to the end of it and you are still looking things up, then I have made you worse rather than better. Shut it. Go and look at the ground in front of you.

HOW IT GOES WRONG

The people who most need to put it down are the people who will not. I know this because they come to the Long Room and I can see it. What they

want is not the rule. It is somebody else to have made the decision. I cannot make it. Every entry in here has a failure printed underneath it for that reason. I put them there so that nobody could read this book as a set of instructions. A great many people are going to read it as a set of instructions anyway. So the last thing, and then I will stop. Everything above is what worked in one valley, for one set of people, against one winter after another, and the sample is made entirely of survivors. Take what is useful. Argue with the rest of it. Write down what you learn, put a name and a date on it, and bring it here. If it is better than what is in this book, then this book will be shorter next year. That is the only outcome I have ever wanted from it.

A Note on the Sources

Eleven hundred notebooks came into the Long Room between the second autumn and this one. Most arrived in bin bags, brought by people who had cleared a house and could not bring themselves to burn the paper in it. Ninety-nine rules came out. It is worth saying plainly what that number is made of, because a manual that will not show its working is asking you to bet your life on a stranger's memory.

Fifty-one are verbatim. The rule is on the page in the hand of the person named, and I have the page. Where the wording is odd I have left it odd. Bea Winslow wrote in the imperative because she had spent forty years writing Post Office notices, and her rules read like Post Office notices, and I am not going to improve them.

Twenty-two are verbatim but undated. The rule is in the notebook and the day count is my estimate, taken from what is around it: a weather note, a death, a mention of the frost. Where I have guessed I have guessed low, because people write rules down after the thing that taught them, not before.

Twenty-five are reconstructed. Somebody told me a rule that somebody else had kept, and the second person was dead, and there is no page. I have written those in the words I was given and named the person they belong to. Two of them I have from three separate mouths that never met, which is the strongest evidence in the collection and also, I am aware, exactly what a well travelled invention looks like.

One is mine. It is the last entry in the book and it is signed with this building, because leaving a gap in a manual is worse than filling it in your own name. Fifty-one and twenty-two and twenty-five and one is ninety-nine, and if that arithmetic ever stops working then somebody has edited this without telling me.

On Idris Pyle. Four rules in this book are attributed to a man who by his own account was a confidence trickster in Sheffield before all this, and who by the account of four other people was a driving instructor. His rules are the best writing in the collection. They are also, two of them, wrong in ways I can demonstrate, and one of those is a rule for taking food off a group of strangers dressed up as a rule for joining one. I have kept all four and marked the two. If you want a single reason to distrust every attribution in this book, it is that the most quotable man in it is the one I am least sure existed. Two of the notebooks in his hand are in different hands.

On the neat ones. Four rules arrived already famous, passed mouth to mouth across three counties, attached to no notebook and no body. They are the four best sentences in the manual. That is the tell. A rule that has been polished by three hundred retellings is a rule that has been optimised for being repeated, which is not the same property as being true, and the entries say so where I could establish it and admit it where I could not.

On the dead. Thirty-six people have rules in this book, and the index gives each of them with the entries they contributed. Two hundred and six are named somewhere in these pages once you count the ones who appear inside an entry rather than at the head of one, and of those, a hundred and seventy-one are dead, twenty-nine are alive as of the first frost, and six I have no word of either way. I have been asked more than once to take the day counts out, on the grounds that they make the book frightening to read. They stay in. A rule is worth what it cost, and the day count is the price on the tag.

On what is missing. There is nothing in here about children under four, because the two people who knew most about that both died in the

second summer and I will not reconstruct what I do not understand. There is nothing about the coast, which is a different apocalypse with different rules, and Hana Sørensen has a notebook of her own and is welcome to it. There is nothing about London.

If you have a notebook, the Long Room is on the corner where the bank was. Bring it. If the rule in it is better than the rule in here, the rule in here comes out.

Index of Names

A

Adeyemi, Femi 50, 57
Adjei, Pearl 34
Anand, Kit 32

B

Beeston wall, the 14
Bello, Terrance 40
Bramall, Isaac 44, 64

C

Cattermole, Bill 30

D

Doherty, Fen 29, 45, 56, 78

E

Errai, Yusuf 13, 20, 65, 88, 92

F

Ferrier, Duncan 46, 55

H

Hatherleigh, Rob 12, 21, 29, 61, 74, 83

K

Kaminski, Rosa 43, 52, 82, 93
Kite, Jonah 19, 54
Kowal, Marta 64, 95

L

Lawal, Ade 62
Long Room, the 100
Lunt, Maggie 35, 47, 63, 79, 94

M

Mackay, Erin 52
Moray, Sister Agnes 25, 79, 89, 94

O

Okafor, Nell 18, 83
Okon, Marcus 9, 21, 41, 73
Okonjo, Ivy 61, 76, 98
Ordish, Wes 86, 99

P

Prentice, Oz 11, 41

Prydderch, Gwen 32, 60

Pyle, Idris 19, 55, 72, 76

Q

Quaye, Solomon 44, 58, 85, 99

R

Raghunathan, Priya 8, 23, 31, 53, 66,
84

Rahman, Nadia 24, 67

Reith, Callum 22, 35, 75, 89

S

Semenova, Lila 33, 42

Skipton radio log, the 24

Sørensen, Hana 96

T

Tait, Ewan 10, 28, 87

Threlfall, Sam 13

Trent, Joyce 15

V

Vellacott, Ruth 77

Vrba, Tomas 11, 73, 97

W

Winslow, Bea 9, 47, 51, 67, 86



99 Ways to Survive a Zombie Apocalypse

Three hundred and fifteen entries across twenty-one chapters,
drawn from a hundred and fifty-one sources.

Set in EB Garamond, cut by Octavio Pardo
after the types of Claude Garamont and Robert Granjon.

